

SEASONAL LANDSLIDE INVENTORY FOR NER - 2022



North Eastern Regional Node for Disaster Risk Reduction (NERDRR)
NORTH EASTERN SPACE APPLICATIONS CENTRE
GOVERNMENT OF INDIA, DEPARTMENT OF SPACE,

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12	Abstract	The study was taken up as an ongoing process under NERDRR programme of NESAC, Umiam. It is basically a compendium of all reported landslide incidents collected from newspapers and online media reports. A total of 145 landslide incidents have been documented for the different states of the NER for the year 2022, majority of which have been triggered by rainfall. Further, a total of 41 casualties have been reported out of which 10 are from Arunachal Pradesh, 11 are from Assam, 8 from Manipur and 12 are from Meghalaya. In addition, collapse of road-sections of National/State Highways is reported from all the states of NER. This report also contains information on 24 hrs total rainfall prior to each event. However, this report is not inclusive of every landslide events that has occurred during the year 2022. Some of these locations are verified in terms of precise latitude and longitude, either through field survey, photographs and/or post-dated imagery. Accordingly, codification has been assigned to all. The systematic inventory thus compiled will help in carrying out future landslide research. Further this will help in assessing the risk in terms of yearly loss of life and property, disruption of transportation network, frequency of landslide and magnitude of landslide incidents in an area and the NER as a whole.		
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PREFACE

North Eastern Regional node for Disaster Risk Reduction (NER-DRR) has been set up at NESAC, Umiam in the year 2011 to provide scientific solutions towards disaster management to the entire region. Almost entire NER consisting of 8 states, namely, Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim and Tripura is susceptible to landslides except the plain areas of Assam, Manipur and Tripura. Physiographically NER is a part of Extra-Peninsula of India and of geologically recent origin, having exposed rocks from Archaean to Quaternary ages. This diverse setup makes the terrain susceptible to landslides. While, intense and/or prolonged monsoonal rainfall and seismic activities are the immediate trigger, anthropogenic factors have a role in accelerating the issue. Hence, the need for immediate and systematic cataloguing of landslide incidents is realized by creating a geospatial database highlighting the landslide locations with respect to nearby settlements and transportation network after the event is reported.

It is expected that, the systematic inventory thus compiled will help in carrying out future landslide research. Further, visualization of the same in spatial context will help in assessing the risk in terms of yearly loss of life and property, disruption of transportation network, frequency and magnitude of landslide incidents in an area and NER as a whole.

DPD, NER-DRR

CONTENTS

PREFACE	v
1. INTRODUCTION	1
2.SOURCES OF DATA	2
2.1LANDSLIDE DATA.....	2
2.1.1NEWSPAPERS:.....	3
2.1.2ONLINE MEDIA:.....	3
2.2SATELLITE DATA.....	4
2.3 RAINFALL DATA	4
3. CHARACTERISITICS OF THE LANDSLIDE INVENTORY	5
4. METHODOLOGY.....	7
4.1 LANDSLIDE POINT AND POLYGON EXTRACTION	7
4.2 AUTOMATIC RAINFALL DATA EXTRACTION	8
5. LANDSLIDE INVENTORY IN DIFFERENT PARTS OF THE NER.....	9
5.1ARUNACHAL PRADESH	9
5.2ASSAM	11
5.3MANIPUR.....	14
5.4MEGHALAYA.....	16
5.5MIZORAM	18
5.6SIKKIM.....	20
6.ANNEXURES	22
ANNEXURE-I: LANDSLIDE INCIDENT REPORTS	22
ANNEXURE-II: NEWSPAPER (CLIPPINGS & DIGITAL ARCHIVE)	45
ANNEXURE-III: ONLINE MEDIA.....	47
ANNEXURE-IV: INVENTORY ATTRIBUTES	48
ANNEXURE-V: CODE SAMPLE.....	49

LIST OF FIGURES

Figure 1: Sentinel 1A and 2C	4
Figure 2: GPM IMERG Late Run Product.....	4
Figure 3: Flowchart for landslide polygon extraction	7
Figure 4: Flowchart for automatic extraction of rainfall	8

LIST OF TABLES

Table 1: Sources of landslide data for different states of NER	2
Table 2: State wise list of newspapers reporting landslide incidents in NER	3
Table 3: Summary of landslide incidents in Arunachal Pradesh.....	9
Table 4: Summary of landslide incidents in Assam.....	12
Table 5: Summary of landslide incidents in Manipur	14
Table 6: Summary of landslide incidents in Meghalaya	16
Table 7: Summary of landslide incidents in Mizoram.....	18
Table 8: Summary of landslide incidents in Sikkim.....	20

LIST OF MAPS

Map 1: Landslide Incidents in the NER, 2022	6
Map 2: Landslide Incidents in Arunachal Pradesh, 2022	10
Map 3: Landslide Incidents in Assam, 2022.....	13
Map 4: Landslide Incidents in Manipur 2022	15
Map 5: Landslide Incidents in Meghalaya, 2022.....	17
Map 6: Landslide Incidents in Mizoram, 2022.....	19
Map7: Landslide Incidents in Sikkim, 2022.....	21

1. INTRODUCTION

The term "landslide" describes a wide variety of processes that result in the downward and outward movement of slope forming materials composed of natural rock, soils, artificial fills or combinations of these material, (Verns, 1958). According to Geological Survey of India (GSI) reports, 427 k sq. km area accounting 12.6% of the Indian landmass is landslide prone and this contributes app. 8% of global landslide fatalities. Hence, the people living in the North Eastern Region of India are very familiar with both the magnitude and the frequency with which landslides wreak havoc each year. NASA's Landslide Hazard Assessment for Situational Awareness Model (LHASA) classifies Landslide Susceptibility in large parts of North East India under categories 4 and 5 (in a scale from 0 to 5).

Under the purview of North Eastern Regional Node for Disaster Risk Reduction (NERDRR), a need for immediate cataloguing of landslide incidents and creation of a Geospatial database has been realized by highlighting the landslide locations with respect to nearby settlements and transportation network after report of any landslide incident. The systematic inventory thus compiled will help in carrying out future landslide research. Further this will help in assessing State/area wise risk in terms of yearly loss of life and property, disruption of transportation network, frequency of landslide and magnitude of landslide incidents for the whole of NER. This brief report provides information on reported landslide incidences pan NER collected from various sources. However, this may not be inclusive of all the events that have triggered during 2022 monsoon season.

2. SOURCES OF DATA

2.1 LANDSLIDE DATA

Unlike other disasters like earthquakes, so far landslide incidents are not instrumentally recorded and reported anywhere immediately. So, immediate cataloguing of landslide incidents has been done mainly from newspapers and online/social media. Subsequently, landslide locations are verified in terms of precise latitude and longitude either through field survey or from post-dated satellite/UAV imagery as and when available. Accordingly, all codification is assigned. The sources of data contributing to this inventory are classified into three categories:

1. Newspapers and Media
2. Field Visit
3. Satellite Image

Due care is taken to minimise the confusion regarding date of occurrences and precise locations of reported events.

Source	Arunachal Pradesh	Assam	Manipur	Meghalaya	Mizoram	Sikkim	Nagaland	Tripura
Total Landslide Events	5	78	1	35	24	2	-	-
Newspapers & Media	5	5	1	7	1	2	-	-
Field Visit	-	43	-	28	23	-	-	-
Satellite Image	-	30	-	-	-	-	-	-

Table 1: Sources of landslide data for different states of NER

2.1.1 NEWSPAPERS:

Newspapers are the primary sources of information that record landslide incidents in NER. Hence, leading daily newspapers distributed in the different states of NER were thoroughly reviewed (Annexure II). In addition, since online media plays a significant role in reporting disaster incidents. Therefore, data collected from newspapers were supplemented with information acquired from online media.

State wise list of newspapers that are referred for 2022 landslide incidents in NER are provided in Table 2.

Newspaper and Media	Assam	Arunachal Pradesh	Manipur	Meghalaya	Mizoram	Nagaland	Sikkim
The Shillong Times	✓	✓		✓			
The Assam Tribune	✓						
The Sangai Express			✓				
The Sikkim Express							✓
Eastern Mirror Nagaland						✓	
The Morung Express						✓	
Youtube Clips		✓	✓	✓		✓	

Table 2: State wise list of newspapers reporting landslide incidents in NER

2.1.2 ONLINE MEDIA:

In the context of this inventory, online media includes all the small local/regional news stations that only have an online presence. With the ubiquitous accessibility of internet, often landslide incidents photographed/recorded by people end up on different social media platforms. Certain community outreach groups also provide information about recent disasters on such sites. Finally, the army provides updates on twitter whenever they help in clearing landslide debris to facilitate movement.

Examples of the same used in this inventory are provided in the Annexure-III.

2.2 SATELLITE DATA

Satellite imagery available post event or post monsoons are used to extract landslide areas through polygon features to validate positional accuracy of the landslide locations. Freely available imagery from SENTINEL 2 Multispectral Instrument (MSI) is mainly used for extraction and updation of landslide features. Details of the Level 1C and the Level 2A products of SENTINEL 2 MSI used for this study is given below.

The [Level-1C](#) product is composed of 100x100 km² tiles (ortho-images in UTM/WGS84 projection). The Level-1C product results from using a [Digital Elevation Model \(DEM\)](#) to project the image in cartographic geometry. Per-pixel radiometric measurements are provided in Top Of Atmosphere (TOA) reflectances along with the parameters to transform them into radiances. Level-1C products are resampled with a constant Ground Sampling Distance (GSD) of 10, 20 and 60 m depending on the native resolution of the different spectral bands. In Level-1C products, pixel coordinates refer to the upper left corner of the pixel.

Level-1C products will additionally include [Cloud Masks](#) and [ECMWF](#) data (total column of ozone, total column of water vapour and mean sea level pressure).

The [Level-2A](#) product provides Bottom Of Atmosphere (BOA) reflectance images derived from the associated Level-1C products. Each Level-2A product is composed of 100x100 km² tiles in cartographic geometry (UTM/WGS84 projection).

Level-2A products are systematically generated at the ground segment over Europe since March 2018, and the production was extended to global in December 2018.

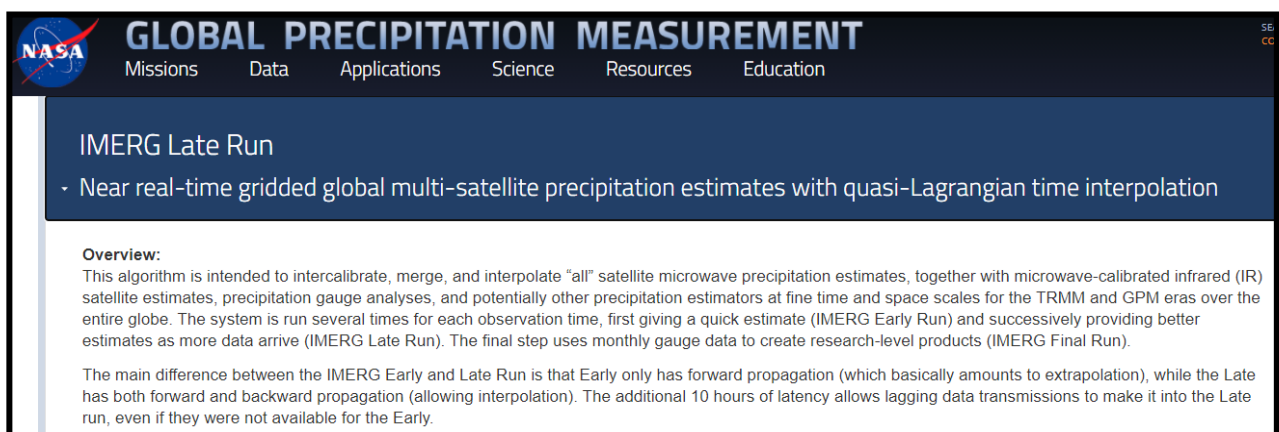
Source: ESA Website

Figure 1: Sentinel 1C and 2A

Both Level 1C and 2A products were used with the Ground Sampling Distance (GSD) of 10m.

2.3 RAINFALL DATA

Rainfall data from NASA's Global Precipitation Mission (GPM) has been used, considering the seamless availability of rainfall data for the entire NER. 'IMERG LATE RUN' product of spatial resolution - 0.1 degrees/~10km has been used.



The screenshot shows the NASA Global Precipitation Measurement (GPM) website. The header includes the NASA logo and navigation links: Missions, Data, Applications, Science, Resources, and Education. The main content area is titled 'IMERG Late Run' and describes it as 'Near real-time gridded global multi-satellite precipitation estimates with quasi-Lagrangian time interpolation'. An 'Overview' section explains that the algorithm intercalibrates, merges, and interpolates satellite microwave precipitation estimates with microwave-calibrated infrared (IR) satellite estimates and precipitation gauge analyses. It notes that the system is run several times for each observation time, first giving a quick estimate (IMERG Early Run) and then providing better estimates as more data arrive (IMERG Late Run). The final step uses monthly gauge data to create research-level products (IMERG Final Run). A concluding paragraph states that the main difference between the IMERG Early and Late Run is that Early only has forward propagation (basically amounts to extrapolation), while the Late has both forward and backward propagation (allowing interpolation). The additional 10 hours of latency allows lagging data transmissions to make it into the Late run, even if they were not available for the Early.

Figure 2: GPM IMERG Late Run Product

3. CHARACTERISTICS OF THE LANDSLIDE INVENTORY

The inventory is seasonal in nature and contains events that are triggered by 2022 monsoonal rainfall.

A total of 145 landslide incidents have been documented in the different states of NER. 41 casualties are reported out of which 10 are from Arunachal Pradesh, 11 from Assam, 8 from Manipur and 12 from Meghalaya. In addition, collapses of road-sections along National/State Highways have also been reported from the states of the NER. Base and other ancillary information were used to correlate the inventory database is of 1:50,000 scale.

This database reflects the landslide locations immediately after an event is reported in newspaper, online/offline media, preferably within 24hrs of reporting, so depending upon the availability of precise location information the inventory database is grouped into following 2 categories: -

- **TYPE 1 (Represented by Green colour)**

These are the events, for which precise latitude/longitude information is available. Such events are generally validated from UAV/Satellite Imagery and/or Field Visits. Post-dated satellite imageries are used for validation purpose only where spatial resolution of the image and size and exposure condition of the landslide makes it possible.

- **TYPE 2 (Represented by Yellow colour)**

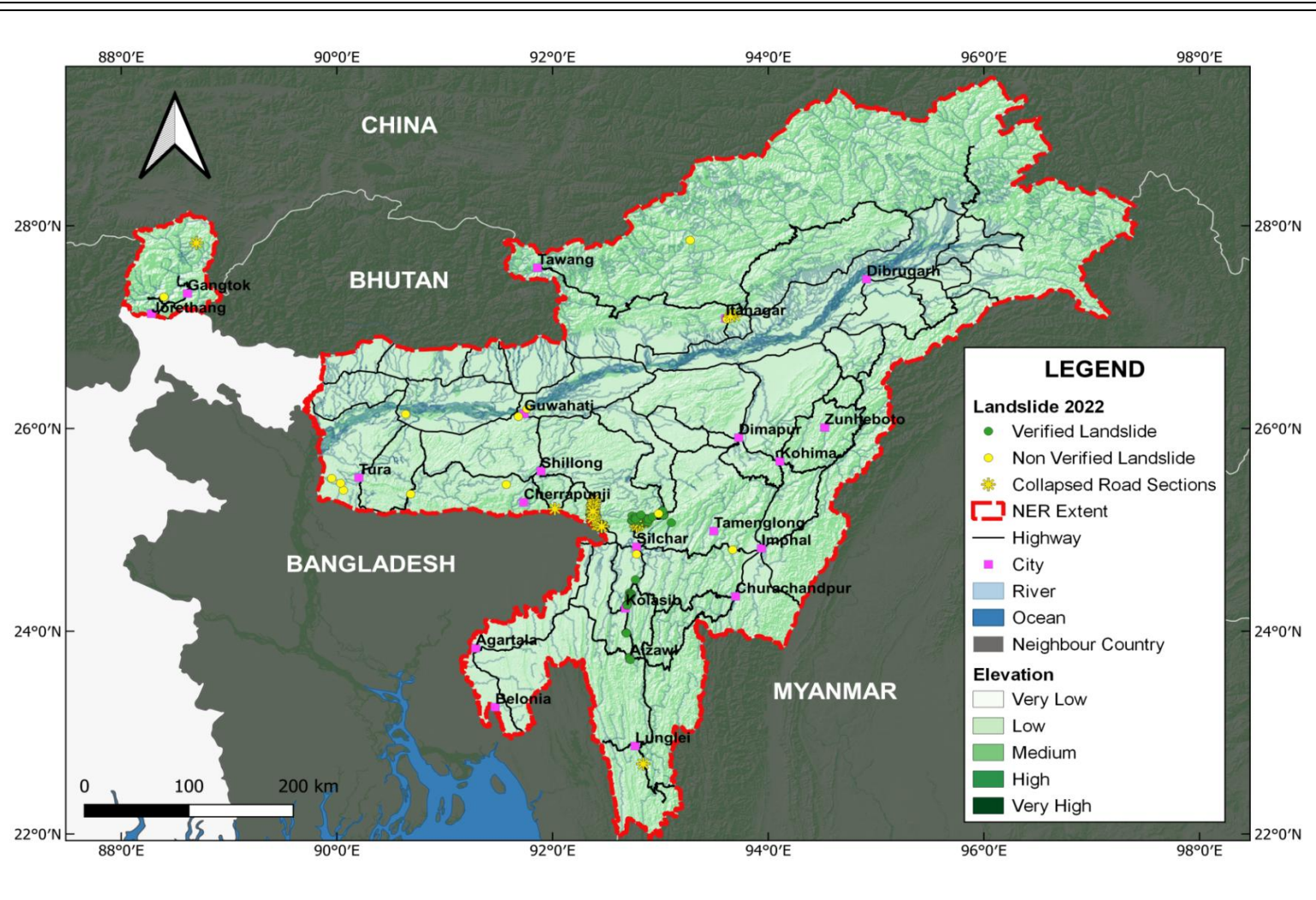
These are the events, for which precise latitude/longitude information not available rather is referred with respect to nearby village/settlement or locality name by newspapers/media reports. In most of the case these slides are of smaller dimension and could not be visualized using freely available Sentinel 2 data of 10m spatial resolution. However, as and when needed the same may be validated by using post-dated high-resolution satellite data.

- A 3rd category (Type 3) includes areas where effect of landslide on road sections along National Highways/State Highways is reported, which has affected transportation. Separate symbology is used to distinguish these locations. However, challenge associated for validation of these locations is that, road sections usually get restored or repaired at the earliest possible time thus only a small window is left for validation purpose either through

a field visit or using satellite imagery. Revisit status of particular satellite further complicates the situation. However, as and when needed the same may be validated by using post-dated satellite data of appropriate resolution.

All the landslide locations also have standard attribute information, in terms of unique ID, location, date/time/type of events, casualty information, 24hrs rainfall prior to the event etc. (Annexure IV). Due care is been taken to validate and convert all the event locations to Type 1 at the earliest.

Landslide Locations 2022, NER, India



Map 1: Landslide Incidents in the NER, 2022

4. METHODOLOGY

4.1 LANDSLIDE POINT AND POLYGON EXTRACTION

For preparation of Seasonal Landslide Inventory post monsoon for the year 2022, a thorough review of articles relevant to landslide incidents from all the major Daily Newspapers in NER was carried out.

The location of all landslide incidents was derived from newspapers and an initial database of landslide locations was prepared in the form of a Point Shapefile.

Of all the landslide incidents, all those events which were determined large enough to be visualized on Sentinel 2 imagery were extracted and a 10km buffer was created around them (for ambiguity resolution in geolocation).

Finally using Pre and Post Event Satellite Imagery, landslide polygons within the 10km buffer wherever visible, were digitized.

The flowchart for the methodology is displayed in Figure 4 and the relevant code is provided in the Annexure V.

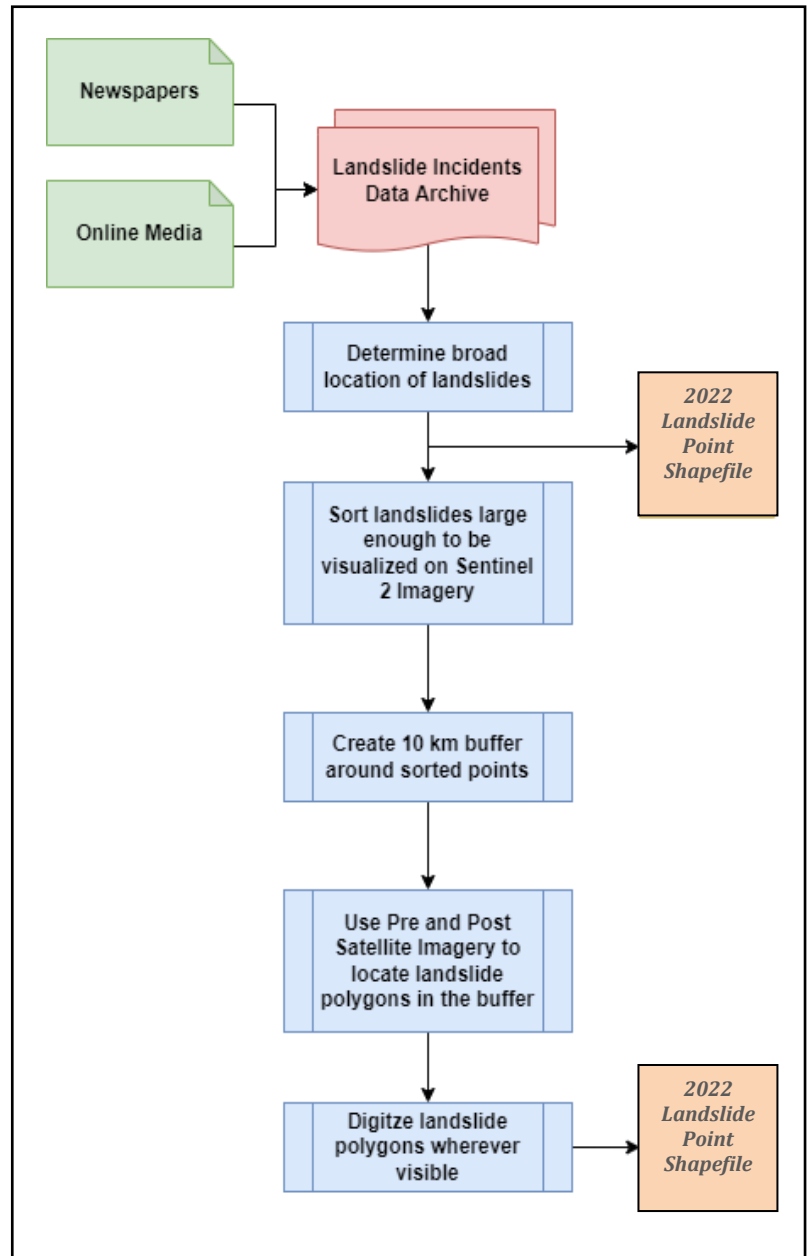


Figure 3: Flowchart for landslide polygon extraction

4.2 AUTOMATIC RAINFALL DATA EXTRACTION

From the list of landslides events, latitude, longitude and date of event are used to extract the corresponding rainfall data using our created automation pipeline. This automation pipeline will first download the required rainfall data from specific sources (Currently GPM data supported) and pre-process them to get hourly precipitation and cumulative precipitation of each event and finally generate the rainfall intensity graph for each event.

The flowchart for the methodology behind Automatic Rainfall Data Extraction is shown in Figure 5.

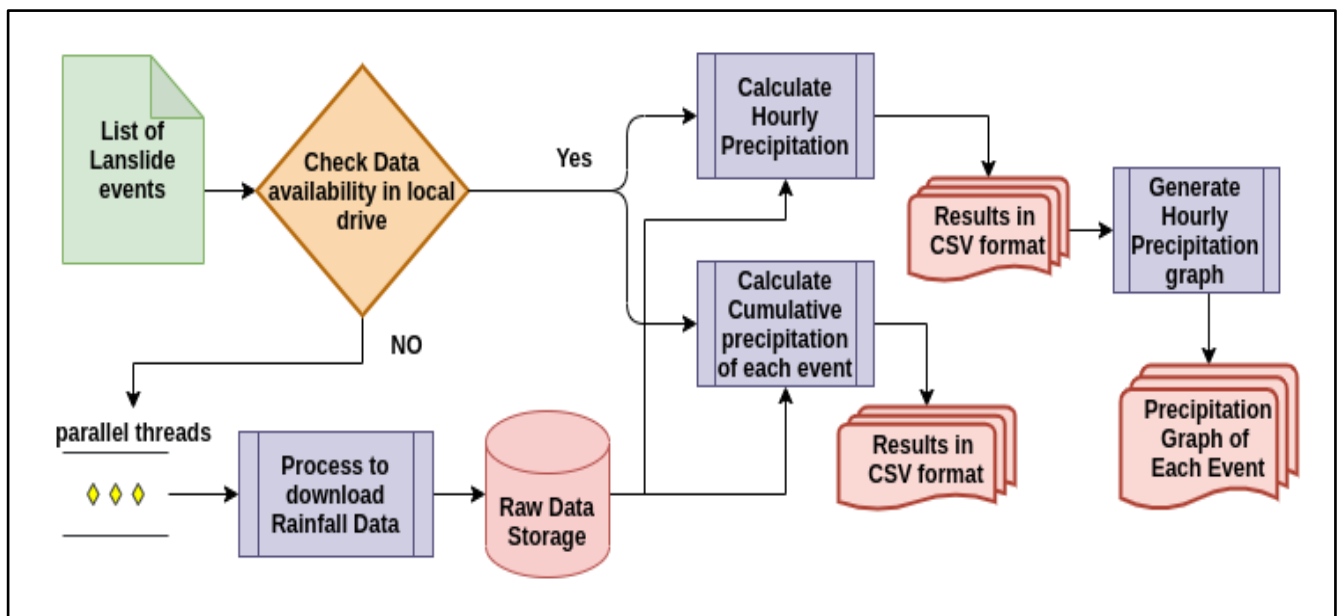


Figure 4: Flowchart for automatic extraction of rainfall

5. LANDSLIDE INVENTORY IN DIFFERENT PARTS OF THE NER

In this section of the report, a very brief summary of landslide incidents in the different states of the NER is provided. Detailed information regarding each landslide incident is provided in the annexure.

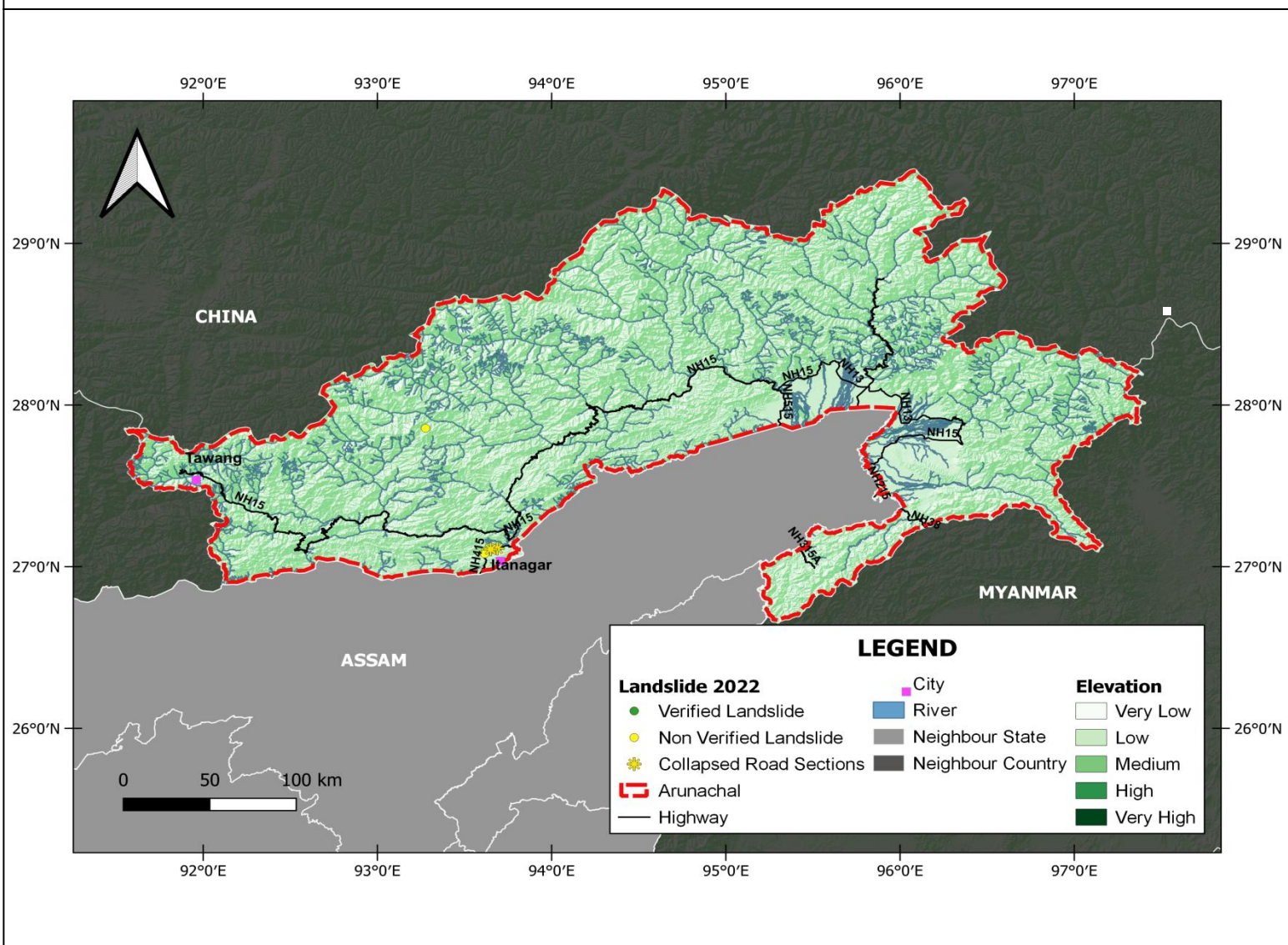
5.1 ARUNACHAL PRADESH

- A total of 5 landslide incidents recorded in Arunachal Pradesh.
- The first landslide incident for the year was recorded on 19th April 2022 and the last event was recorded on 19th June 2022.
- A total of 10 casualties occurred in Arunachal Pradesh in 2022
- NH-415 and Yupia-Doimukh road were some of the highways/road sections that were affected in the state due to landslides.

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Newspapers&Media/2022/Arunachal Pradesh/106	Newspapers&Media	Arunachal Pradesh	Papum Pare	19-06-2022	2	1
2	Newspapers&Media/2022/Arunachal Pradesh/107	Newspapers&Media	Arunachal Pradesh	Papum Pare	16-05-2022	2	4
3	Newspapers&Media/2022/Arunachal Pradesh/108	Newspapers&Media	Arunachal Pradesh	Papum Pare	16-05-2022	2	2
4	Newspapers&Media/2022/Arunachal Pradesh/109	Newspapers&Media	Arunachal Pradesh	Papum Pare	19-06-2022	2	0
5	Newspapers&Media/2022/Arunachal Pradesh/110	Newspapers&Media	Arunachal Pradesh	Papum Pare	19-04-2022	2	3

Table 3: Summary of landslide incidents in Arunachal Pradesh

Landslide Locations 2022, Arunachal Pradesh



Map 2: Landslide Incidents in Arunachal Pradesh, 2022

5.2 ASSAM

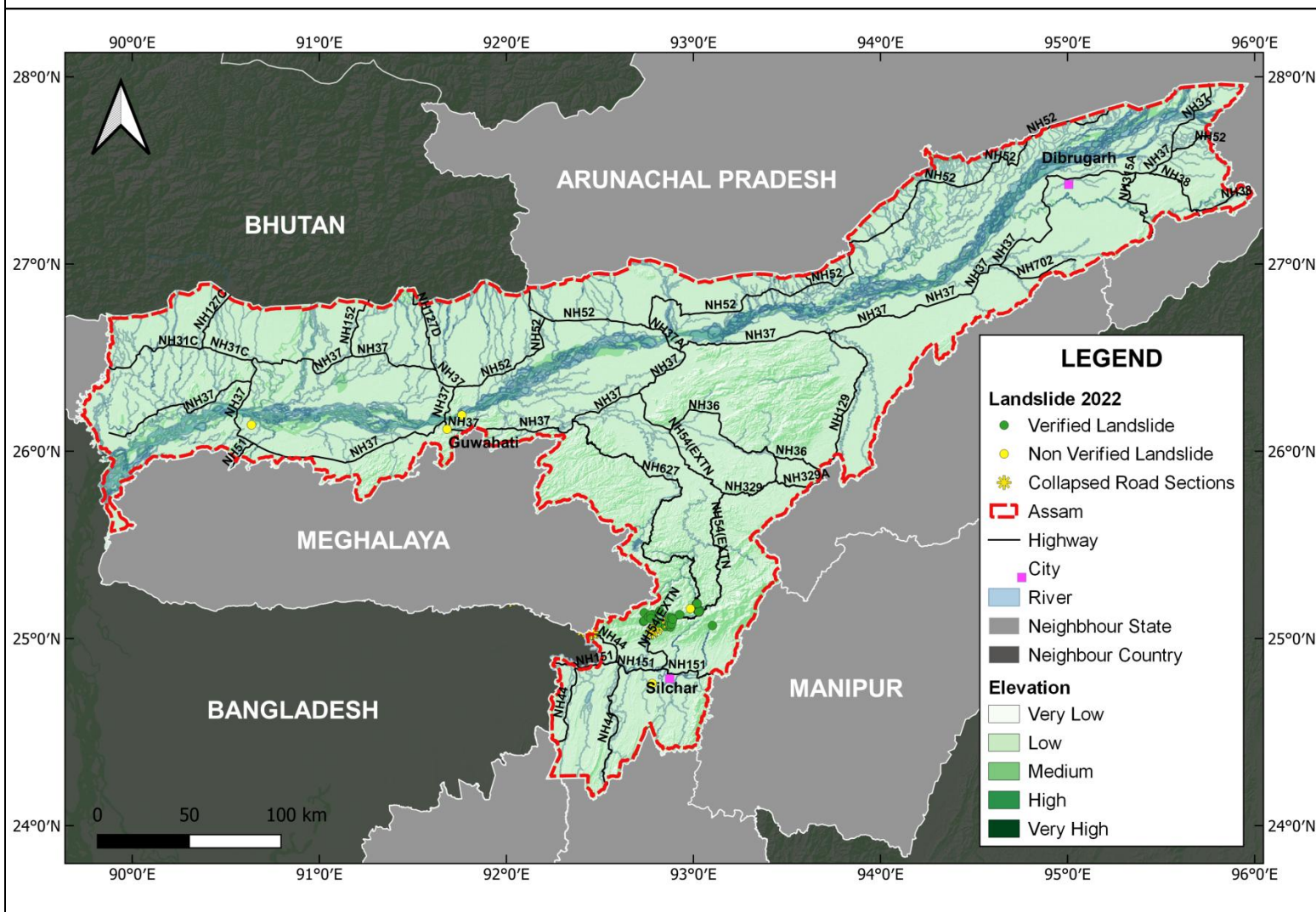
- A total of 78 landslide incidents recorded in Assam.
- The first landslide incident for the year was recorded on 10th April 2022 and the last event was recorded on 17th June 2022.
- A total of 11 casualties occurred in Assam in 2022
- NH 27 and NH 6 highways/road sections that were affected in the state due to landslides.

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	field_visit/2022/Assam/29	field_visit	Assam	Cachar	10-04-2022	1	0
2	field_visit/2022/Assam/30	field_visit	Assam	Cachar	10-04-2022	1	0
3	field_visit/2022/Assam/31	field_visit	Assam	Cachar	10-04-2022	1	0
4	field_visit/2022/Assam/32	field_visit	Assam	Cachar	10-04-2022	1	0
5	field_visit/2022/Assam/33	field_visit	Assam	Cachar	10-04-2022	1	0
6	field_visit/2022/Assam/34	field_visit	Assam	Cachar	10-04-2022	1	0
7	satellite_image/2022/Assam/35	satellite image	Assam	Cachar	15-05-2022	1	0
8	satellite_image/2022/Assam/36	satellite image	Assam	Cachar	15-05-2022	1	0
9	satellite_image/2022/Assam/37	satellite image	Assam	Cachar	15-05-2022	1	0
10	satellite_image/2022/Assam/38	satellite image	Assam	Cachar	15-05-2022	1	0
11	satellite_image/2022/Assam/39	satellite image	Assam	Cachar	15-05-2022	1	0
12	satellite_image/2022/Assam/40	satellite image	Assam	Cachar	15-05-2022	1	0
13	satellite_image/2022/Assam/41	satellite image	Assam	Cachar	15-05-2022	1	0
14	satellite_image/2022/Assam/42	satellite image	Assam	Cachar	15-05-2022	1	0
15	satellite_image/2022/Assam/43	satellite image	Assam	Cachar	15-05-2022	1	0
16	satellite_image/2022/Assam/44	satellite image	Assam	Cachar	15-05-2022	1	0
17	satellite_image/2022/Assam/45	satellite image	Assam	Cachar	15-05-2022	1	0
18	satellite_image/2022/Assam/46	satellite image	Assam	Cachar	15-05-2022	1	0
19	satellite_image/2022/Assam/47	satellite image	Assam	Dima Hasao	15-05-2022	1	0
20	satellite_image/2022/Assam/48	satellite image	Assam	Dima Hasao	15-05-2022	1	0
21	satellite_image/2022/Assam/49	satellite image	Assam	Dima Hasao	15-05-2022	1	0
22	satellite_image/2022/Assam/50	satellite image	Assam	Dima Hasao	15-05-2022	1	0
23	satellite_image/2022/Assam/51	satellite image	Assam	Dima Hasao	15-05-2022	1	0
24	satellite_image/2022/Assam/52	satellite image	Assam	Dima Hasao	15-05-2022	1	0
25	satellite_image/2022/Assam/53	satellite image	Assam	Dima Hasao	15-05-2022	1	0
26	satellite_image/2022/Assam/54	satellite image	Assam	Dima Hasao	15-05-2022	1	0
27	satellite_image/2022/Assam/55	satellite image	Assam	Dima Hasao	15-05-2022	1	0
28	satellite_image/2022/Assam/56	satellite image	Assam	Dima Hasao	15-05-2022	1	0
29	satellite_image/2022/Assam/57	satellite image	Assam	Dima Hasao	15-05-2022	1	0
30	satellite_image/2022/Assam/58	satellite image	Assam	Dima Hasao	15-05-2022	1	0

31	satellite_image/2022/Assam/59	satellite image	Assam	Dima Hasao	15-05-2022	1	0
32	satellite_image/2022/Assam/60	satellite image	Assam	Dima Hasao	15-05-2022	1	0
33	satellite_image/2022/Assam/61	satellite image	Assam	Dima Hasao	15-05-2022	1	0
34	satellite_image/2022/Assam/62	satellite image	Assam	Dima Hasao	15-05-2022	1	0
35	satellite_image/2022/Assam/63	satellite image	Assam	Dima Hasao	15-05-2022	1	0
36	satellite_image/2022/Assam/64	satellite image	Assam	Dima Hasao	15-05-2022	1	0
37	field_visit/2022/Assam/65	field_visit	Assam	Dima Hasao	15-05-2022	1	0
38	field_visit/2022/Assam/66	field_visit	Assam	Dima Hasao	15-05-2022	1	0
39	field_visit/2022/Assam/67	field_visit	Assam	Dima Hasao	15-05-2022	1	0
40	field_visit/2022/Assam/68	field_visit	Assam	Dima Hasao	15-05-2022	1	0
41	field_visit/2022/Assam/69	field_visit	Assam	Dima Hasao	15-05-2022	1	0
42	field_visit/2022/Assam/70	field_visit	Assam	Dima Hasao	15-05-2022	1	0
43	field_visit/2022/Assam/71	field_visit	Assam	Dima Hasao	15-05-2022	1	0
44	field_visit/2022/Assam/72	field_visit	Assam	Dima Hasao	15-05-2022	1	0
45	field_visit/2022/Assam/73	field_visit	Assam	Dima Hasao	15-05-2022	1	0
46	field_visit/2022/Assam/74	field_visit	Assam	Dima Hasao	15-05-2022	1	0
47	field_visit/2022/Assam/75	field_visit	Assam	Dima Hasao	15-05-2022	1	0
48	field_visit/2022/Assam/76	field_visit	Assam	Dima Hasao	15-05-2022	1	0
49	field_visit/2022/Assam/77	field_visit	Assam	Dima Hasao	15-05-2022	1	0
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51	field_visit/2022/Assam/79	field_visit	Assam	Dima Hasao	15-05-2022	1	0
52	field_visit/2022/Assam/80	field_visit	Assam	Dima Hasao	15-05-2022	1	0
53	field_visit/2022/Assam/81	field_visit	Assam	Dima Hasao	15-05-2022	1	0
54	field_visit/2022/Assam/82	field_visit	Assam	Dima Hasao	15-05-2022	1	0
55	field_visit/2022/Assam/83	field_visit	Assam	Dima Hasao	15-05-2022	1	0
56	field_visit/2022/Assam/84	field_visit	Assam	Dima Hasao	15-05-2022	1	0
57	field_visit/2022/Assam/85	field_visit	Assam	Dima Hasao	15-05-2022	1	0
58	field_visit/2022/Assam/86	field_visit	Assam	Dima Hasao	15-05-2022	1	0
59	field_visit/2022/Assam/87	field_visit	Assam	Dima Hasao	15-05-2022	1	0
60	field_visit/2022/Assam/88	field_visit	Assam	Dima Hasao	15-05-2022	1	0
61	field_visit/2022/Assam/89	field_visit	Assam	Dima Hasao	15-05-2022	1	0
62	field_visit/2022/Assam/90	field_visit	Assam	Dima Hasao	15-05-2022	1	0
63	field_visit/2022/Assam/91	field_visit	Assam	Dima Hasao	15-05-2022	1	0
64	field_visit/2022/Assam/92	field_visit	Assam	Dima Hasao	15-05-2022	1	0
65	field_visit/2022/Assam/93	field_visit	Assam	Dima Hasao	15-05-2022	1	0
66	field_visit/2022/Assam/94	field_visit	Assam	Dima Hasao	15-05-2022	1	0
67	field_visit/2022/Assam/95	field_visit	Assam	Dima Hasao	15-05-2022	1	0
68	field_visit/2022/Assam/96	field_visit	Assam	Dima Hasao	15-05-2022	1	0
69	field_visit/2022/Assam/97	field_visit	Assam	Dima Hasao	15-05-2022	1	0
70	field_visit/2022/Assam/98	field_visit	Assam	Dima Hasao	15-05-2022	1	0
71	field_visit/2022/Assam/99	field_visit	Assam	Dima Hasao	15-05-2022	1	0
72	field_visit/2022/Assam/100	field_visit	Assam	Dima Hasao	15-05-2022	1	0
73	field_visit/2022/Assam/101	field_visit	Assam	Dima Hasao	15-05-2022	1	0
74	Newspapers&Media/2022/Assam/102	Newspapers&Media	Assam	Dima Hasao	14-05-2022	2	3
75	Newspapers&Media/2022/Assam/104	Newspapers&Media	Assam	Cachar	18-06-2022	2	2
76	Newspapers&Media/2022/Assam/105	Newspapers&Media	Assam	Kamrup Metro	19-06-2022	2	0
77	Newspapers&Media/2022/Assam/111	Newspapers&Media	Assam	Kurung Kumey	12-06-2022	2	4
78	Newspapers&Media/2022/Assam/112	Newspapers&Media	Assam	Gopalpara	17-06-2022	2	2

Table 4: Summary of landslide incidents in Assam

Landslide Locations 2022, Assam



Map 3: Landslide Incidents in Assam, 2022

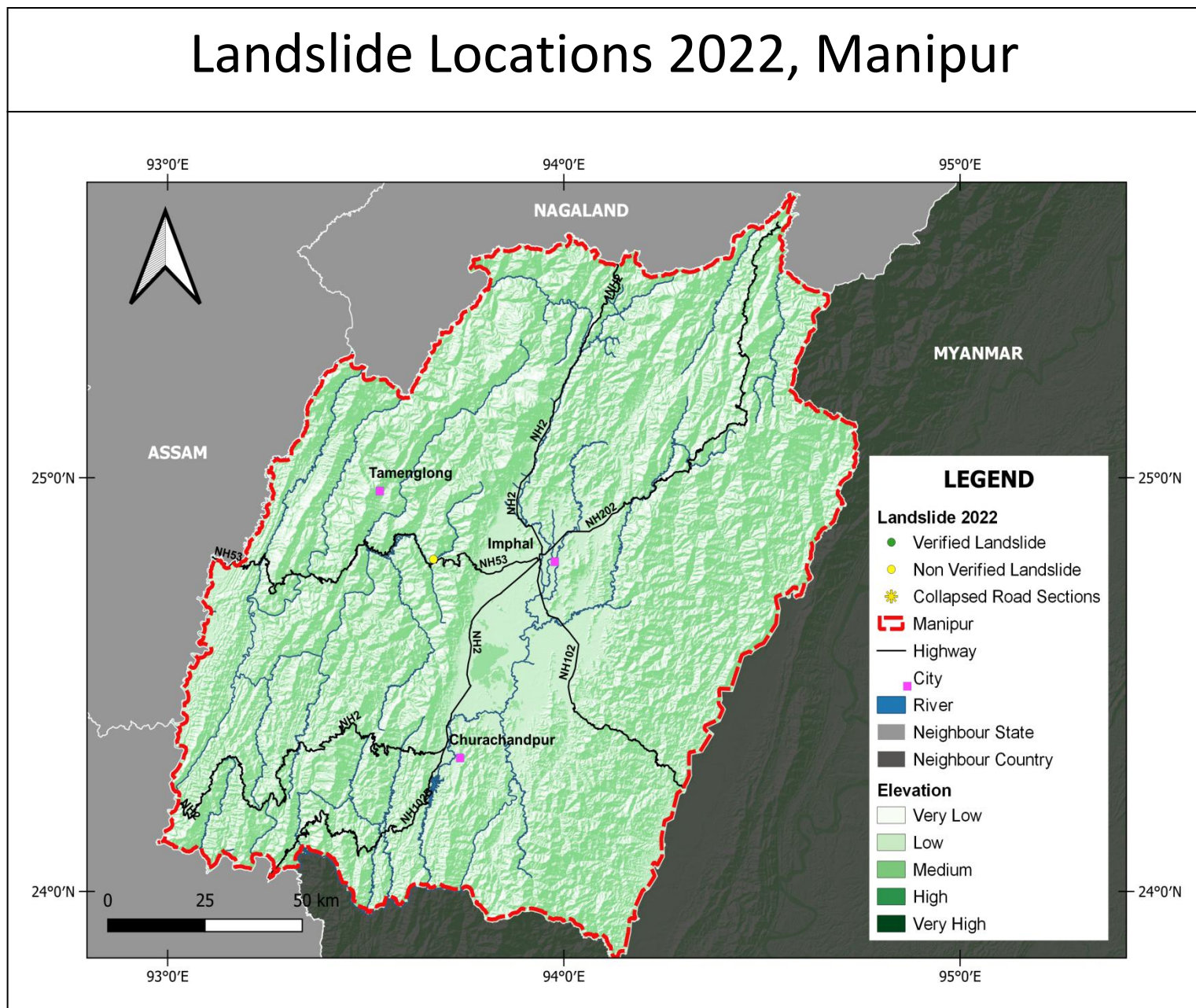
5.3 MANIPUR

- 1 landslide incident recorded in Manipur.
- The landslide incident for the year was recorded on 30th June 2022.
- A total of 8 casualties occurred in Manipur in 2022.
- No highways/road sections that were affected in the state due to landslides.

Sr No	Landslide ID	Source	State	District	Date	Type	casualty
1	Newspapers&Media/2022/Manipur/120	Newspapers&Media	Manipur	Noney	30-06-2022	2	8

Table 5: Summary of landslide incidents in Manipur

Landslide Locations 2022, Manipur



Map 2: Landslide Incidents in Manipur 2022

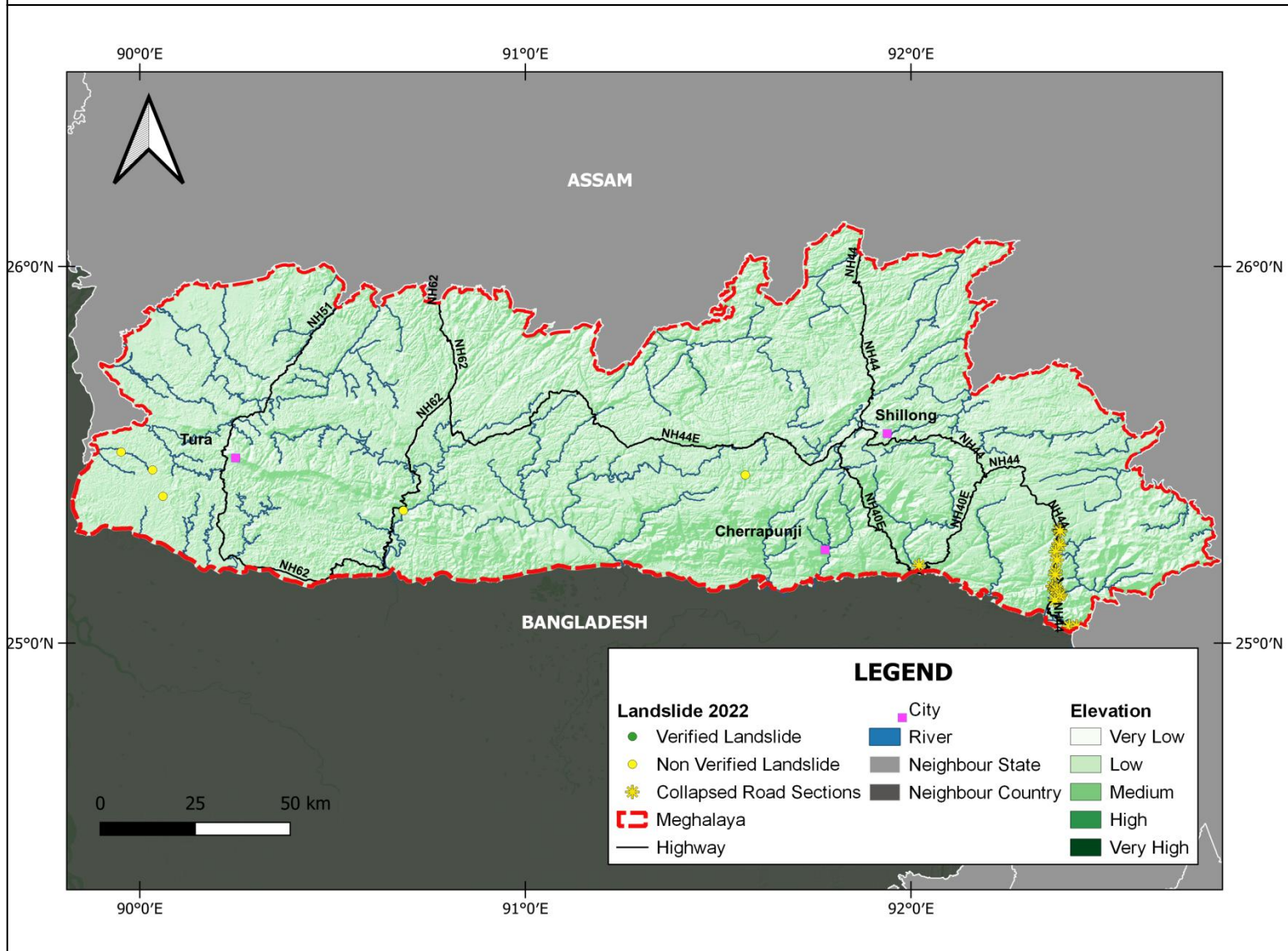
5.4 MEGHALAYA

- A total of 35 landslide incidents recorded in Meghalaya.
- The first landslide incident for the year was recorded on 13th April 2022 and the last event was recorded on 17th June 2022.
- 12 casualties were reported from Meghalaya in 2022.
- NH 6 and Darrang Shnongpdeng Road were some of the highways/road sections that were affected in the state due to landslides.

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	field_visit/2022/Meghalaya/1	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
2	field_visit/2022/Meghalaya/2	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
3	field_visit/2022/Meghalaya/3	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
4	field_visit/2022/Meghalaya/4	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
5	field_visit/2022/Meghalaya/5	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
6	field_visit/2022/Meghalaya/6	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
7	field_visit/2022/Meghalaya/7	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
8	field_visit/2022/Meghalaya/8	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
9	field_visit/2022/Meghalaya/9	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
10	field_visit/2022/Meghalaya/10	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
11	field_visit/2022/Meghalaya/11	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
12	field_visit/2022/Meghalaya/12	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
13	field_visit/2022/Meghalaya/13	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
14	field_visit/2022/Meghalaya/14	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
15	field_visit/2022/Meghalaya/15	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
16	field_visit/2022/Meghalaya/16	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
17	field_visit/2022/Meghalaya/17	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
18	field_visit/2022/Meghalaya/18	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
19	field_visit/2022/Meghalaya/19	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
20	field_visit/2022/Meghalaya/20	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
21	field_visit/2022/Meghalaya/21	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
22	field_visit/2022/Meghalaya/22	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
23	field_visit/2022/Meghalaya/23	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
24	field_visit/2022/Meghalaya/24	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
25	field_visit/2022/Meghalaya/25	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
26	field_visit/2022/Meghalaya/26	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
27	field_visit/2022/Meghalaya/27	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
28	field_visit/2022/Meghalaya/28	field_visit	Meghalaya	Jaintia Hills	13-04-2022	1	0
29	Newspapers&Media/2022/Meghalaya/113	Newspapers&Media	Meghalaya	West Garo Hills	09-06-2022	2	3
30	Newspapers&Media/2022/Meghalaya/114	Newspapers&Media	Meghalaya	South West Garo Hills	09-06-2022	2	1
31	Newspapers&Media/2022/Meghalaya/115	Newspapers&Media	Meghalaya	East Jaintia Hills	16-06-2020	2	0
32	Newspapers&Media/2022/Meghalaya/116	Newspapers&Media	Meghalaya	East Khasi Hills	16-06-2022	2	4
33	Newspapers&Media/2022/Meghalaya/117	Newspapers&Media	Meghalaya	West Jaintia Hills	17-06-2022	2	0
34	Newspapers&Media/2022/Meghalaya/118	Newspapers&Media	Meghalaya	South Garo Hills	17-06-2022	2	3
35	Newspapers&Media/2022/Meghalaya/119	Newspapers&Media	Meghalaya	South Garo Hills	17-06-2022	2	1

Table 6: Summary of landslide incidents in Meghalaya

Landslide Locations 2022, Meghalaya



Map 5: Landslide Incidents in Meghalaya 2022

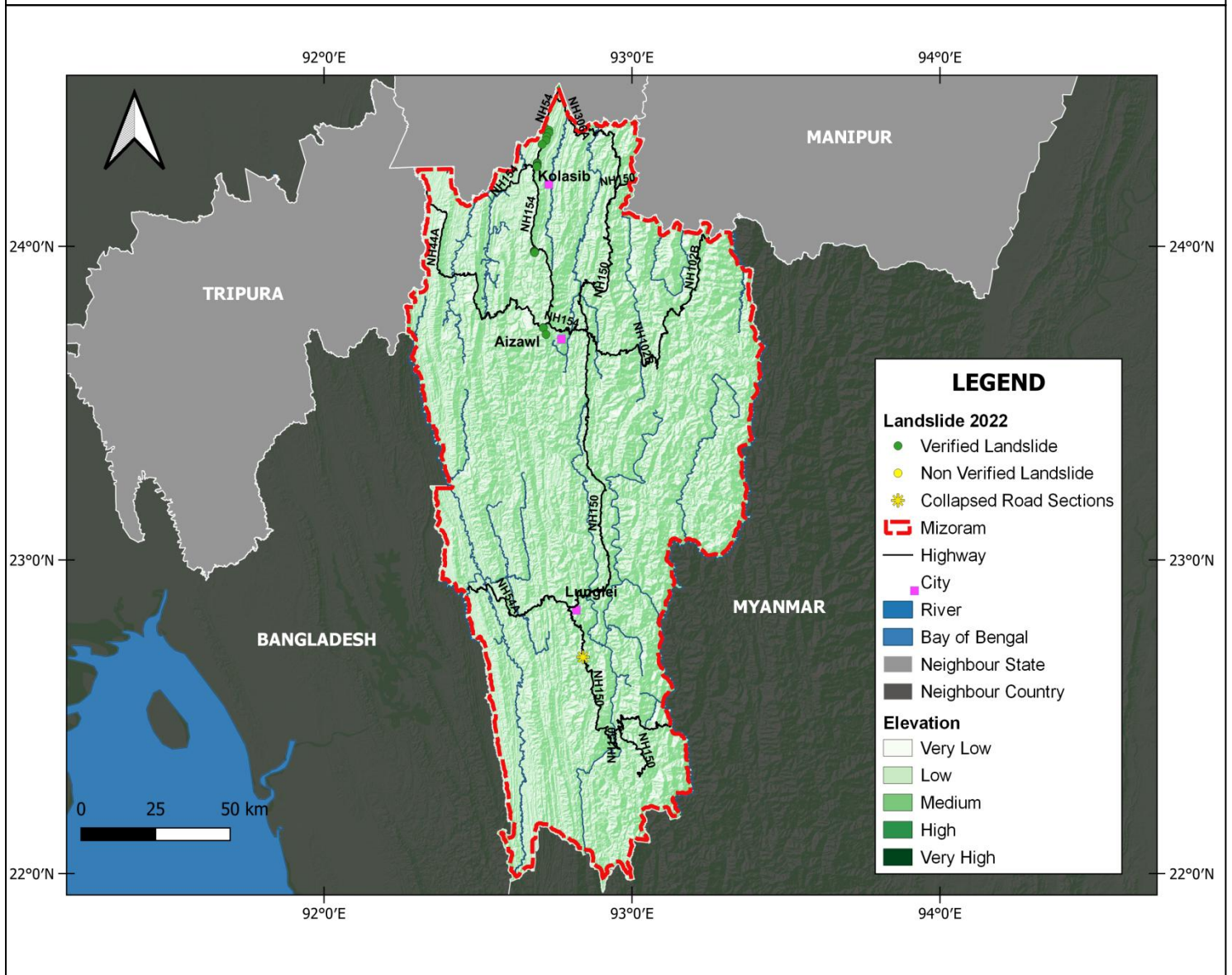
5.5 MIZORAM

- 24 landslide incidents recorded in Mizoram.
- The first landslide incident was recorded on 19th June 2022.
- No casualties occurred in Mizoram in 2022.
- Tawipui Road was affected by landslides in Mizoram in 2022.

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Newspapers&Media/2022/Mizoram/103	Newspapers&Media	Mizoram	Lunglei	19-06-2022	2	0
2	field_visit/2022/Mizoram/123	field_visit	Mizoram	Kolasib	24-07-2022	1	0
3	field_visit/2022/Mizoram/124	field_visit	Mizoram	Aizawl	24-07-2022	1	0
4	field_visit/2022/Mizoram/125	field_visit	Mizoram	Aizawl	24-07-2022	1	0
5	field_visit/2022/Mizoram/126	field_visit	Mizoram	Aizawl	24-07-2022	1	0
6	field_visit/2022/Mizoram/127	field_visit	Mizoram	Aizawl	24-07-2022	1	0
7	field_visit/2022/Mizoram/128	field_visit	Mizoram	Aizawl	24-07-2022	1	0
8	field_visit/2022/Mizoram/129	field_visit	Mizoram	Kolasib	24-07-2022	1	0
9	field_visit/2022/Mizoram/130	field_visit	Mizoram	Kolasib	24-07-2022	1	0
10	field_visit/2022/Mizoram/131	field_visit	Mizoram	Kolasib	24-07-2022	1	0
11	field_visit/2022/Mizoram/132	field_visit	Mizoram	Serchhip	24-07-2022	1	0
12	field_visit/2022/Mizoram/133	field_visit	Mizoram	Serchhip	24-07-2022	1	0
13	field_visit/2022/Mizoram/134	field_visit	Mizoram	Serchhip	24-07-2022	1	0
14	field_visit/2022/Mizoram/135	field_visit	Mizoram	Serchhip	24-07-2022	1	0
15	field_visit/2022/Mizoram/136	field_visit	Mizoram	Serchhip	24-07-2022	1	0
16	field_visit/2022/Mizoram/137	field_visit	Mizoram	Serchhip	24-07-2022	1	0
17	field_visit/2022/Mizoram/138	field_visit	Mizoram	Kolasib	24-07-2022	1	0
18	field_visit/2022/Mizoram/139	field_visit	Mizoram	Kolasib	24-07-2022	1	0
19	field_visit/2022/Mizoram/140	field_visit	Mizoram	Kolasib	24-07-2022	1	0
20	field_visit/2022/Mizoram/141	field_visit	Mizoram	Kolasib	24-07-2022	1	0
21	field_visit/2022/Mizoram/142	field_visit	Mizoram	Kolasib	24-07-2022	1	0
22	field_visit/2022/Mizoram/143	field_visit	Mizoram	Aizawl	26-07-2022	1	0
23	field_visit/2022/Mizoram/144	field_visit	Mizoram	Aizawl	26-07-2022	1	0
24	field_visit/2022/Mizoram/145	field_visit	Mizoram	Aizawl	26-07-2022	1	0

Table 7: Summary of landslide incidents in Mizoram

Landslide Locations 2022, Mizoram



Map 3: Landslide Incidents in Mizoram, 2022

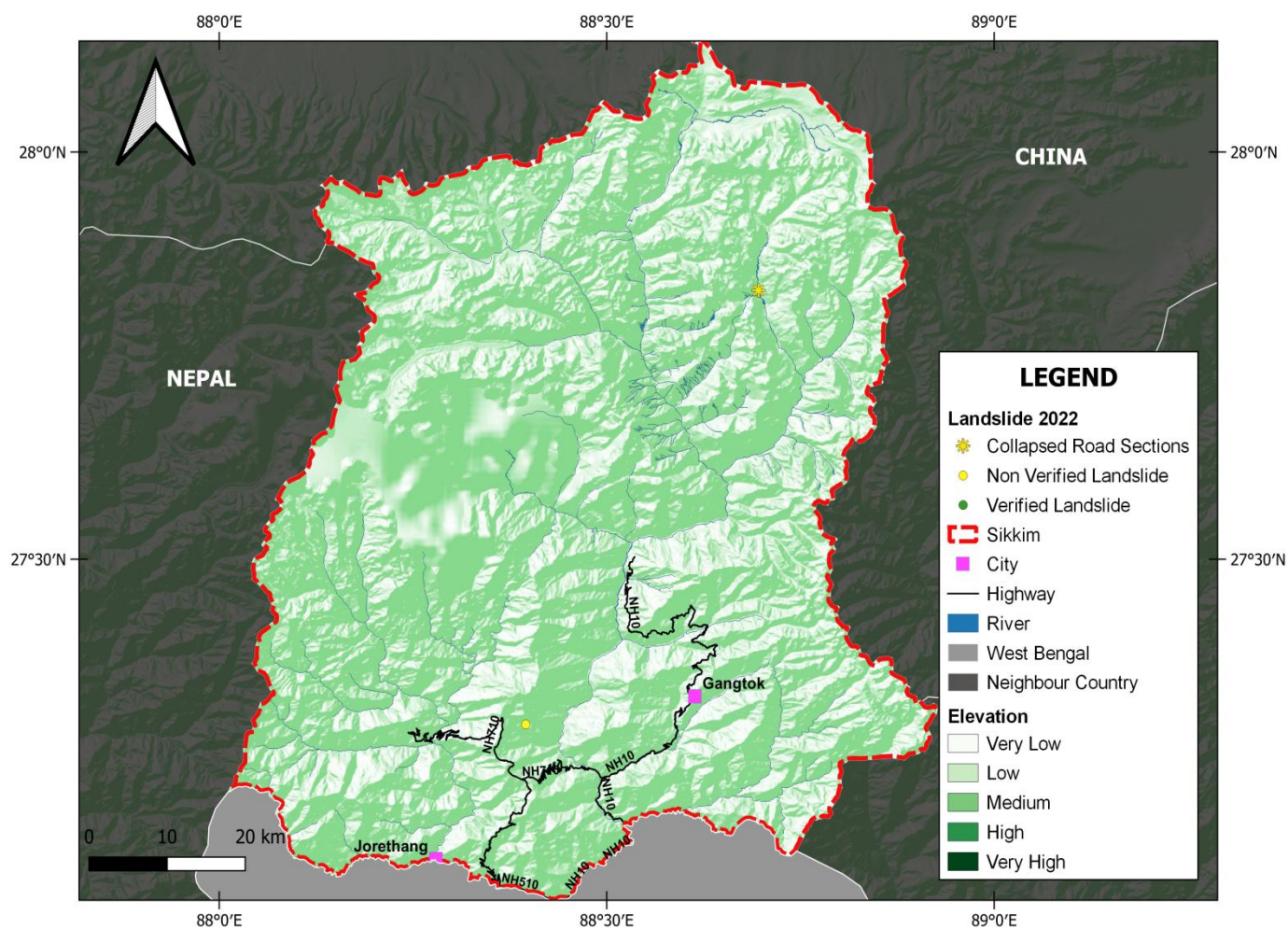
5.6 SIKKIM

- 2 landslide incidents recorded in Sikkim.
- The landslide incident for the year was recorded on 31th August 2022.
- No casualties were reported from Sikkim in 2022.
- Yumthang Valley Road affected by landslide.

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Newspapers&Media/2022/Sikkim/121	Newspapers&Media	Sikkim	North Sikkim	31-08-2022	2	0
2	Newspapers&Media/2022/Sikkim/122	Newspapers&Media	Sikkim	South Sikkim	04-11-2022	2	0

Table 8: Summary of landslide incidents in Sikkim

Landslide Locations 2022, Sikkim



Map 7: Landslide Incidents in Sikkim, 2022

ANNEXURES

ANNEXURE-I: LANDSLIDE INCIDENT REPORTS

MEGHALAYA

Slide ID	field_visit/2022/Meghalaya/1	Slide ID	field_visit/2022/Meghalaya/2
Source	field_visit	Source	field_visit
Slide No	1	Slide No	2
State	Meghalaya	State	Meghalaya
District	Jaintia Hills	District	Jaintia Hills
Name	Byndihati-Umtyra_Road_Creep	Name	Byndihati-Umtyra_1
Location	Between Byndihati and Umtyra	Location	Between Byndihati and Umtyra
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.300713	Lat(dd)	25.299032
Lon(dd)	92.3835	Lon(dd)	92.386284
Date of Event	13-04-2022	Date of Event	13-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	12.557893	Rainfall(mm)	12.557893

Slide ID	field_visit/2022/Meghalaya/3	Slide ID	field_visit/2022/Meghalaya/4
Source	field_visit	Source	field_visit
Slide No	3	Slide No	4
State	Meghalaya	State	Meghalaya
District	Jaintia Hills	District	Jaintia Hills
Name	Byndihati-Umtyra_2	Name	Byndihati-Umtyra_3
Location	Between Byndihati and Umtyra	Location	Between Byndihati and Umtyra
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.298933	Lat(dd)	25.29857
Lon(dd)	92.38648	Lon(dd)	92.386894
Date of Event	13-04-2022	Date of Event	13-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	12.557893	Rainfall(mm)	12.557893

Slide ID	field_visit/2022/Meghalaya/5	Slide ID	field_visit/2022/Meghalaya/6
Source	field_visit	Source	field_visit
Slide No	5	Slide No	6
State	Meghalaya	State	Meghalaya
District	Jaintia Hills	District	Jaintia Hills
Name	Byndihati-Umtyra_4	Name	Byndihati-Umtyra_5
Location	Between Byndihati and Umtyra	Location	Between Byndihati and Umtyra
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.29842	Lat(dd)	25.298416
Lon(dd)	92.386955	Lon(dd)	92.386986
Date of Event	13-04-2022	Date of Event	13-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	12.557893	Rainfall(mm)	0
Slide ID	field_visit/2022/Meghalaya/7	Slide ID	field_visit/2022/Meghalaya/8
Source	field_visit	Source	field_visit
Slide No	7	Slide No	8
State	Meghalaya	State	Meghalaya
District	Jaintia Hills	District	Jaintia Hills
Name	Byndihati-Umtyra_6	Name	Byndihati-Umtyra_7
Location	Between Byndihati and Umtyra	Location	Between Byndihati and Umtyra
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.298306	Lat(dd)	25.297895
Lon(dd)	92.386986	Lon(dd)	92.38725
Date of Event	13-04-2022	Date of Event	13-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	0	Rainfall(mm)	0
Slide ID	field_visit/2022/Meghalaya/9	Slide ID	field_visit/2022/Meghalaya/10
Source	field_visit	Source	field_visit
Slide No	9	Slide No	10
State	Meghalaya	State	Meghalaya
District	Jaintia Hills	District	Jaintia Hills
Name	Byndihati-Umtyra_8	Name	Byndihati-Umtyra_9
Location	Between Byndihati and Umtyra	Location	Between Byndihati and Umtyra
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.297684	Lat(dd)	25.297401
Lon(dd)	92.38736	Lon(dd)	92.38735
Date of Event	13-04-2022	Date of Event	13-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	0	Rainfall(mm)	12.557893

Slide ID	field_visit/2022/Meghalaya/12	Slide ID	field_visit/2022/Meghalaya/13
Source	field_visit	Source	field_visit
Slide No	12	Slide No	13
State	Meghalaya	State	Meghalaya
District	Jaintia Hills	District	Jaintia Hills
Name	Nongsning_12	Name	Shiehruphi_1
Location	Near Nongsning	Location	Shiehruphi
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.259485	Lat(dd)	25.242348
Lon(dd)	92.38551	Lon(dd)	92.3757
Date of Event	13-04-2022	Date of Event	13-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	12.557893	Rainfall(mm)	62.07355
Slide ID	field_visit/2022/Meghalaya/14	Slide ID	field_visit/2022/Meghalaya/16
Source	field_visit	Source	field_visit
Slide No	14	Slide No	16
State	Meghalaya	State	Meghalaya
District	Jaintia Hills	District	Jaintia Hills
Name	Thangskai_1	Name	Umlaper_tea_stall_1
Location	near Thangskai	Location	near Raju Tea Stall, near Umlaper
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.220339	Lat(dd)	25.150444
Lon(dd)	92.37896	Lon(dd)	92.36781
Date of Event	13-04-2022	Date of Event	13-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	111.5892	Rainfall(mm)	111.5892
Slide ID	field_visit/2022/Meghalaya/17	Slide ID	field_visit/2022/Meghalaya/18
Source	field_visit	Source	field_visit
Slide No	17	Slide No	18
State	Meghalaya	State	Meghalaya
District	Jaintia Hills	District	Jaintia Hills
Name	Tongseng1	Name	Tongseng2
Location	near Tongseng Village	Location	near Tongseng Village
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.147703	Lat(dd)	25.143984
Lon(dd)	92.37676	Lon(dd)	92.373055
Date of Event	13-04-2022	Date of Event	13-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	111.5892	Rainfall(mm)	111.5892

Slide ID	field_visit/2022/Meghalaya/19	Slide ID	field_visit/2022/Meghalaya/20
Source	field_visit	Source	field_visit
Slide No	19	Slide No	20
State	Meghalaya	State	Meghalaya
District	Jaintia Hills	District	Jaintia Hills
Name	Tongseng3	Name	Tongseng4
Location	near Tongseng Village	Location	near Tongseng Village
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.1438	Lat(dd)	25.144033
Lon(dd)	92.37366	Lon(dd)	92.37443
Date of Event	13-04-2022	Date of Event	13-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	111.5892	Rainfall(mm)	111.5892
Slide ID	field_visit/2022/Meghalaya/21	Slide ID	field_visit/2022/Meghalaya/22
Source	field_visit	Source	field_visit
Slide No	21	Slide No	22
State	Meghalaya	State	Meghalaya
District	Jaintia Hills	District	Jaintia Hills
Name	HP_petrol_pump_tongseng_1	Name	HP_petrol_pump_tongseng_2
Location	behind HP Petrol Pump Tongseng	Location	behind HP Petrol Pump Tongseng
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.14076	Lat(dd)	25.140772
Lon(dd)	92.379005	Lon(dd)	92.37903
Date of Event	13-04-2022	Date of Event	13-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	111.5892	Rainfall(mm)	111.5892
Slide ID	field_visit/2022/Meghalaya/23	Slide ID	field_visit/2022/Meghalaya/24
Source	field_visit	Source	field_visit
Slide No	23	Slide No	24
State	Meghalaya	State	Meghalaya
District	Jaintia Hills	District	Jaintia Hills
Name	HP_petrol_pump_tongseng_3	Name	3km_before_sonapur_1
Location	behind HP Petrol Pump Tongseng	Location	3 km before sonapur
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.140808	Lat(dd)	25.129538
Lon(dd)	92.37898	Lon(dd)	92.388245
Date of Event	13-04-2022	Date of Event	13-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	111.5892	Rainfall(mm)	111.5892

Slide ID	field_visit/2022/Meghalaya/25	Slide ID	field_visit/2022/Meghalaya/26
Source	field_visit	Source	field_visit
Slide No	25	Slide No	26
State	Meghalaya	State	Meghalaya
District	Jaintia Hills	District	Jaintia Hills
Name	Before_Sonapur2	Name	Before_Sonapur3
Location	before sonapur	Location	before sonapur
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.12547	Lat(dd)	25.113762
Lon(dd)	92.38578	Lon(dd)	92.37368
Date of Event	13-04-2022	Date of Event	13-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	111.5892	Rainfall(mm)	111.5892
Slide ID	field_visit/2022/Meghalaya/27	Slide ID	field_visit/2022/Meghalaya/28
Source	field_visit	Source	field_visit
Slide No	27	Slide No	28
State	Meghalaya	State	Meghalaya
District	Jaintia Hills	District	Jaintia Hills
Name	Donsakul_1	Name	Ratacherra_1
Location	near donsakul	Location	near ratacherra
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.045042	Lat(dd)	25.044744
Lon(dd)	92.410835	Lon(dd)	92.43025
Date of Event	13-04-2022	Date of Event	13-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	0	Rainfall(mm)	0
Slide ID	Newspapers&Media/2022/Meghalaya/113	Slide ID	Newspapers&Media/2022/Meghalaya/114
Source	Newspapers&Media	Source	Newspapers&Media
Slide No	113	Slide No	114
State	Meghalaya	State	Meghalaya
District	West Garo Hills	District	South West Garo Hills
Name	Jebalgre_landslide	Name	Betasingh_landslide
Location	Jebalgre Village, Gambegre 45 km from Tura	Location	Samati village, Betasingh block
NH/SH affected		NH/SH affected	
Lat(dd)	25.389677	Lat(dd)	25.507116
Lon(dd)	90.060165	Lon(dd)	89.951546
Date of Event	09-06-2022	Date of Event	09-06-2022
Time of Event		Time of Event	
Type of Event	2	Type of Event	2
Casualty	3	Casualty	1
Rainfall(mm)	66.05802	Rainfall(mm)	98.77895

Slide ID	Newspapers&Media/2022/Meghalaya/115	Slide ID	Newspapers&Media/2022/Meghalaya/116
Source	Newspapers&Media	Source	Newspapers&Media
Slide No	115	Slide No	116
State	Meghalaya	State	Meghalaya
District	East Jaintia Hills	District	East Khasi Hills
Name	Lumshnong_landslide	Name	Laitlarem_landslide
Location	near lumshnong	Location	Laitlarem village under Mawphlang C&RD block
NH/SH affected	NH6	NH/SH affected	
Lat(dd)	25.18166	Lat(dd)	25.44632
Lon(dd)	92.37444	Lon(dd)	91.57084
Date of Event	16-06-2020	Date of Event	16-06-2022
Time of Event		Time of Event	
Type of Event	2	Type of Event	2
Casualty	0	Casualty	4
Rainfall(mm)	152.97546	Rainfall(mm)	191.60959
Slide ID	Newspapers&Media/2022/Meghalaya/117	Slide ID	Newspapers&Media/2022/Meghalaya/118
Source	Newspapers&Media	Source	Newspapers&Media
Slide No	117	Slide No	118
State	Meghalaya	State	Meghalaya
District	West Jaintia Hills	District	South Garo Hills
Name	Darrang_landslide	Name	Bolsalgre_landslide
Location	Darrang Village, Dawki Police Station	Location	Bolsalgre locality, Baghmara
NH/SH affected	Darrang Shnongpdeng Road	NH/SH affected	
Lat(dd)	25.206291	Lat(dd)	25.459652
Lon(dd)	92.02121	Lon(dd)	90.03326
Date of Event	17-06-2022	Date of Event	17-06-2022
Time of Event		Time of Event	
Type of Event	2	Type of Event	2
Casualty	0	Casualty	3
Rainfall(mm)	135.58302	Rainfall(mm)	113.74753
Slide ID	Newspapers&Media/2022/Meghalaya/119		
Source	Newspapers&Media		
Slide No	119		
State	Meghalaya		
District	South Garo Hills		
Name	Siju_landslide		
Location	Siju Village		
NH/SH affected			
Lat(dd)	25.351841		
Lon(dd)	90.68374		
Date of Event	17-06-2022		
Time of Event			
Type of Event	2		
Casualty	1		
Rainfall(mm)	197.52902		

ASSAM

Slide ID	field_visit/2022/Assam/29	Slide ID	field_visit/2022/Assam/30
Source	field_visit	Source	field_visit
Slide No	29	Slide No	30
State	Assam	State	Assam
District	Cachar	District	Cachar
Name	Malidor_meghalaya_1	Name	Malidor_after_bridge_1
Location	Malidor, Meghalaya near bridge	Location	Malidor after crossing bridge
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.033175	Lat(dd)	25.031351
Lon(dd)	92.45981	Lon(dd)	92.463036
Date of Event	10-04-2022	Date of Event	10-04-2022
Time of Event	Midnight	Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	157.13937	Rainfall(mm)	157.13937

Slide ID	field_visit/2022/Assam/31	Slide ID	field_visit/2022/Assam/32
Source	field_visit	Source	field_visit
Slide No	31	Slide No	32
State	Assam	State	Assam
District	Cachar	District	Cachar
Name	Malidor_after_bridge_2	Name	Malidor_Indian_oil_pp1
Location	Malidor after crossing bridge	Location	near Indian Oil Petrol Pump
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.029991	Lat(dd)	25.031443
Lon(dd)	92.464066	Lon(dd)	92.45863
Date of Event	10-04-2022	Date of Event	10-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	157.13937	Rainfall(mm)	157.13937

Slide ID	field_visit/2022/Assam/33	Slide ID	field_visit/2022/Assam/34
Source	field_visit	Source	field_visit
Slide No	33	Slide No	34
State	Assam	State	Assam
District	Cachar	District	Cachar
Name	Malidor_Indian_oil_pp2	Name	Malidor_meghalaya_4
Location	near Indian Oil Petrol Pump	Location	Malidor, Meghalaya near bridge
NH/SH affected	NH 6 (Shillong Bypass)	NH/SH affected	NH 6 (Shillong Bypass)
Lat(dd)	25.03144	Lat(dd)	25.031765
Lon(dd)	92.458496	Lon(dd)	92.455505
Date of Event	10-04-2022	Date of Event	10-04-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	157.13937	Rainfall(mm)	157.13937
Slide ID	satellite_image/2022/Assam/35	Slide ID	satellite_image/2022/Assam/36
Source	satellite image	Source	satellite image
Slide No	35	Slide No	36
State	Assam	State	Assam
District	Cachar	District	Cachar
Name	devi-nala-1	Name	devi-nala-2
Location	devi nala	Location	devi nala
NH/SH affected	NH 27 (lumding silchar road)	NH/SH affected	NH 27 (lumding silchar road)
Lat(dd)	25.03473	Lat(dd)	25.035023
Lon(dd)	92.779236	Lon(dd)	92.78028
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	175.80562	Rainfall(mm)	175.80562
Slide ID	satellite_image/2022/Assam/37	Slide ID	satellite_image/2022/Assam/38
Source	satellite image	Source	satellite image
Slide No	37	Slide No	38
State	Assam	State	Assam
District	Cachar	District	Cachar
Name	bandarkhal_nrsc_south1	Name	bandarkhal_north_1
Location	bandarkhal	Location	bandarkhal
NH/SH affected	NH 27 (lumding silchar road)	NH/SH affected	
Lat(dd)	25.049883	Lat(dd)	25.061815
Lon(dd)	92.80106	Lon(dd)	92.79994
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	140.67575	Rainfall(mm)	159.7697

Slide ID	satellite_image/2022/Assam/39	Slide ID	satellite_image/2022/Assam/40
Source	satellite image	Source	satellite image
Slide No	39	Slide No	40
State	Assam	State	Assam
District	Cachar	District	Cachar
Name	bandarkhal_north_2	Name	bandarkhal_north_east_1
Location	bandarkhal	Location	bandarkhal
NH/SH affected		NH/SH affected	NH 27 (lumding silchar road)
Lat(dd)	25.062498	Lat(dd)	25.066957
Lon(dd)	92.79922	Lon(dd)	92.80956
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	174.04718	Rainfall(mm)	140.67575
Slide ID	satellite_image/2022/Assam/41	Slide ID	satellite_image/2022/Assam/42
Source	satellite image	Source	satellite image
Slide No	41	Slide No	42
State	Assam	State	Assam
District	Cachar	District	Cachar
Name	doloi_chunga_s1	Name	doloi_chunga_s2
Location	doloi chungu	Location	doloi chungu
NH/SH affected		NH/SH affected	
Lat(dd)	25.082067	Lat(dd)	25.065077
Lon(dd)	92.81372	Lon(dd)	92.85103
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	140.67575	Rainfall(mm)	140.67575
Slide ID	satellite_image/2022/Assam/43	Slide ID	satellite_image/2022/Assam/44
Source	satellite image	Source	satellite image
Slide No	43	Slide No	44
State	Assam	State	Assam
District	Cachar	District	Cachar
Name	lodhi_basti1	Name	lodhi_basti2
Location	lodhi basti	Location	lodhi basti
NH/SH affected		NH/SH affected	
Lat(dd)	25.064798	Lat(dd)	25.064266
Lon(dd)	92.8521	Lon(dd)	92.87685
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	140.67575	Rainfall(mm)	140.67575

Slide ID	satellite_image/2022/Assam/45	Slide ID	satellite_image/2022/Assam/46
Source	satellite image	Source	satellite image
Slide No	45	Slide No	46
State	Assam	State	Assam
District	Cachar	District	Cachar
Name	saisi_w1	Name	nachangjol_1
Location	saisi	Location	nachangjol
NH/SH affected		NH/SH affected	
Lat(dd)	25.061062	Lat(dd)	25.07341
Lon(dd)	92.87666	Lon(dd)	92.8663
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	140.67575	Rainfall(mm)	140.67575
Slide ID	satellite_image/2022/Assam/47	Slide ID	satellite_image/2022/Assam/48
Source	satellite image	Source	satellite image
Slide No	47	Slide No	48
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	Maharajpur	Name	bungkhang_sw1
Location	Maharajpur	Location	bungkhang
NH/SH affected	NH 27 (lunding silchar road)	NH/SH affected	
Lat(dd)	25.08474	Lat(dd)	25.08061
Lon(dd)	92.868065	Lon(dd)	92.87005
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	140.67575	Rainfall(mm)	140.67575
Slide ID	satellite_image/2022/Assam/49	Slide ID	satellite_image/2022/Assam/50
Source	satellite image	Source	satellite image
Slide No	49	Slide No	50
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	bungkhang_sw2	Name	bungkhang_3
Location	bungkhang	Location	bungkhang
NH/SH affected		NH/SH affected	
Lat(dd)	25.074862	Lat(dd)	25.078197
Lon(dd)	92.87329	Lon(dd)	92.88341
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	140.67575	Rainfall(mm)	140.67575

Slide ID	satellite_image/2022/Assam/51	Slide ID	satellite_image/2022/Assam/52
Source	satellite image	Source	satellite image
Slide No	51	Slide No	52
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	bungkhang_1	Name	bungkhang_2
Location	bungkhang	Location	bungkhang
NH/SH affected		NH/SH affected	
Lat(dd)	25.091064	Lat(dd)	25.095066
Lon(dd)	92.733284	Lon(dd)	92.737595
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	175.80562	Rainfall(mm)	175.80562
Slide ID	satellite_image/2022/Assam/53	Slide ID	satellite_image/2022/Assam/54
Source	satellite image	Source	satellite image
Slide No	53	Slide No	54
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	doloi_chunga_s3	Name	doloi_chunga_se1
Location	doloi chungu	Location	doloi chungu
NH/SH affected		NH/SH affected	
Lat(dd)	25.137493	Lat(dd)	25.115389
Lon(dd)	92.73741	Lon(dd)	92.75996
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	195.35896	Rainfall(mm)	195.35896
Slide ID	satellite_image/2022/Assam/55	Slide ID	satellite_image/2022/Assam/56
Source	satellite image	Source	satellite image
Slide No	55	Slide No	56
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	nachangjol_2	Name	nachangjol_3
Location	nachangjol	Location	nachangjol
NH/SH affected		NH/SH affected	
Lat(dd)	25.128378	Lat(dd)	25.124657
Lon(dd)	92.78221	Lon(dd)	92.77173
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	195.35896	Rainfall(mm)	195.35896

Slide ID	satellite_image/2022/Assam/57	Slide ID	satellite_image/2022/Assam/58
Source	satellite image	Source	satellite image
Slide No	57	Slide No	58
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	nachangjol_4	Name	new_harangjao1
Location	nachangjol	Location	new harangjao
NH/SH affected		NH/SH affected	
Lat(dd)	25.106842	Lat(dd)	25.111677
Lon(dd)	92.77259	Lon(dd)	92.8712
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	195.35896	Rainfall(mm)	182.28976
Slide ID	satellite_image/2022/Assam/59	Slide ID	satellite_image/2022/Assam/60
Source	satellite image	Source	satellite image
Slide No	59	Slide No	60
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	new_harangjao2	Name	new_harangjao3
Location	new harangjao	Location	new harangjao
NH/SH affected		NH/SH affected	
Lat(dd)	25.110813	Lat(dd)	25.103504
Lon(dd)	92.8845	Lon(dd)	92.88966
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	182.28976	Rainfall(mm)	182.28976
Slide ID	satellite_image/2022/Assam/61	Slide ID	satellite_image/2022/Assam/62
Source	satellite image	Source	satellite image
Slide No	61	Slide No	62
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	harangajao_nw	Name	khomunom
Location	harangjao	Location	khomunom
NH/SH affected		NH/SH affected	
Lat(dd)	25.147371	Lat(dd)	25.069427
Lon(dd)	92.820015	Lon(dd)	93.1018
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	182.28976	Rainfall(mm)	95.41374

Slide ID	satellite_image/2022/Assam/63	Slide ID	satellite_image/2022/Assam/64
Source	satellite image	Source	satellite image
Slide No	63	Slide No	64
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	hill_haflong1	Name	jatinga_lumpur
Location	hill_halflong	Location	jatonga_lumpur
NH/SH affected		NH/SH affected	
Lat(dd)	25.15346	Lat(dd)	25.127455
Lon(dd)	93.0184	Lon(dd)	92.9266
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	182.39934
Slide ID	field_visit/2022/Assam/65	Slide ID	field_visit/2022/Assam/66
Source	field_visit	Source	field_visit
Slide No	65	Slide No	66
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L1	Name	DF1
Location	Near Halflong Railway Station	Location	Near Halflong Railway Station
NH/SH affected		NH/SH affected	
Lat(dd)	25.151909	Lat(dd)	25.149664
Lon(dd)	93.03285	Lon(dd)	93.03088
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583
Slide ID	field_visit/2022/Assam/67	Slide ID	field_visit/2022/Assam/68
Source	field_visit	Source	field_visit
Slide No	67	Slide No	68
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	DF2	Name	L2
Location	Near Halflong Railway Station	Location	Near Halflong Railway Station
NH/SH affected		NH/SH affected	
Lat(dd)	25.149872	Lat(dd)	25.148912
Lon(dd)	93.03178	Lon(dd)	93.031586
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583

Slide ID	field_visit/2022/Assam/69	Slide ID	field_visit/2022/Assam/70
Source	field_visit	Source	field_visit
Slide No	69	Slide No	70
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L3	Name	L8
Location	Near Halflong Railway Station	Location	Near Halflong Railway Station
NH/SH affected		NH/SH affected	
Lat(dd)	25.14752	Lat(dd)	25.145393
Lon(dd)	93.03081	Lon(dd)	93.028824
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583
Slide ID	field_visit/2022/Assam/71	Slide ID	field_visit/2022/Assam/72
Source	field_visit	Source	field_visit
Slide No	71	Slide No	72
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L5	Name	L7
Location	Near Halflong Railway Station	Location	Near Halflong Railway Station
NH/SH affected		NH/SH affected	
Lat(dd)	25.145468	Lat(dd)	25.144587
Lon(dd)	93.03013	Lon(dd)	93.02707
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583
Slide ID	field_visit/2022/Assam/73	Slide ID	field_visit/2022/Assam/74
Source	field_visit	Source	field_visit
Slide No	73	Slide No	74
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L6	Name	DF6
Location	Near Halflong Railway Station	Location	Near Halflong Railway Station
NH/SH affected		NH/SH affected	
Lat(dd)	25.141546	Lat(dd)	25.143412
Lon(dd)	93.03085	Lon(dd)	93.03172
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583

Slide ID	field_visit/2022/Assam/75	Slide ID	field_visit/2022/Assam/76
Source	field_visit	Source	field_visit
Slide No	75	Slide No	76
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L4	Name	DF5
Location	Near Halflong Railway Station	Location	Near Halflong Railway Station
NH/SH affected		NH/SH affected	
Lat(dd)	25.145105	Lat(dd)	25.145308
Lon(dd)	93.03218	Lon(dd)	93.02946
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583
Slide ID	field_visit/2022/Assam/77	Slide ID	field_visit/2022/Assam/78
Source	field_visit	Source	field_visit
Slide No	77	Slide No	78
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	DF4	Name	L9a
Location	Near Halflong Railway Station	Location	Near Halflong Railway Station
NH/SH affected		NH/SH affected	
Lat(dd)	25.144354	Lat(dd)	25.148092
Lon(dd)	93.02944	Lon(dd)	93.0336
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583
Slide ID	field_visit/2022/Assam/79	Slide ID	field_visit/2022/Assam/80
Source	field_visit	Source	field_visit
Slide No	79	Slide No	80
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L9b	Name	L10
Location	Near Halflong Railway Station	Location	Near Kendri Vidyalaya, Halflong
NH/SH affected		NH/SH affected	
Lat(dd)	25.14794	Lat(dd)	25.186506
Lon(dd)	93.03361	Lon(dd)	93.01687
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583

Slide ID	field_visit/2022/Assam/81	Slide ID	field_visit/2022/Assam/82
Source	field_visit	Source	field_visit
Slide No	81	Slide No	82
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L11	Name	L12
Location	Near Kendri Vidyalaya, Halflong	Location	Near Kendri Vidyalaya, Halflong
NH/SH affected		NH/SH affected	
Lat(dd)	25.18653	Lat(dd)	25.187185
Lon(dd)	93.01751	Lon(dd)	93.01716
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583
Slide ID	field_visit/2022/Assam/83	Slide ID	field_visit/2022/Assam/84
Source	field_visit	Source	field_visit
Slide No	83	Slide No	84
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L13	Name	L14
Location	Near Kendri Vidyalaya, Halflong	Location	Near Kendri Vidyalaya, Halflong
NH/SH affected		NH/SH affected	
Lat(dd)	25.18724	Lat(dd)	25.188253
Lon(dd)	93.0178	Lon(dd)	93.01809
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583
Slide ID	field_visit/2022/Assam/85	Slide ID	field_visit/2022/Assam/86
Source	field_visit	Source	field_visit
Slide No	85	Slide No	86
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L15	Name	L16
Location	Near Kendri Vidyalaya, Halflong	Location	Near Kendri Vidyalaya, Halflong
NH/SH affected		NH/SH affected	
Lat(dd)	25.187881	Lat(dd)	25.187809
Lon(dd)	93.01832	Lon(dd)	93.018524
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583

Slide ID	field_visit/2022/Assam/87	Slide ID	field_visit/2022/Assam/88
Source	field_visit	Source	field_visit
Slide No	87	Slide No	88
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L17	Name	L18
Location	Near Kendri Vidyalaya, Halflong	Location	Near Kendri Vidyalaya, Halflong
NH/SH affected		NH/SH affected	
Lat(dd)	25.187763	Lat(dd)	25.187649
Lon(dd)	93.018745	Lon(dd)	93.01909
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583
Slide ID	field_visit/2022/Assam/89	Slide ID	field_visit/2022/Assam/90
Source	field_visit	Source	field_visit
Slide No	89	Slide No	90
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L22	Name	L19
Location	Near Kendri Vidyalaya, Halflong	Location	Near Kendri Vidyalaya, Halflong
NH/SH affected		NH/SH affected	
Lat(dd)	25.186731	Lat(dd)	25.186766
Lon(dd)	93.0194	Lon(dd)	93.01914
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583
Slide ID	field_visit/2022/Assam/91	Slide ID	field_visit/2022/Assam/92
Source	field_visit	Source	field_visit
Slide No	91	Slide No	92
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L27	Name	L28
Location	Near Kendri Vidyalaya, Halflong	Location	Near Kendri Vidyalaya, Halflong
NH/SH affected		NH/SH affected	
Lat(dd)	25.186045	Lat(dd)	25.185911
Lon(dd)	93.01877	Lon(dd)	93.01879
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583

Slide ID	field_visit/2022/Assam/93	Slide ID	field_visit/2022/Assam/94
Source	field_visit	Source	field_visit
Slide No	93	Slide No	94
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L29	Name	DF3
Location	Near Kendri Vidyalaya, Halflong	Location	Near Kendri Vidyalaya, Halflong
NH/SH affected		NH/SH affected	
Lat(dd)	25.186026	Lat(dd)	25.143646
Lon(dd)	93.019104	Lon(dd)	93.02876
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583
Slide ID	field_visit/2022/Assam/95	Slide ID	field_visit/2022/Assam/96
Source	field_visit	Source	field_visit
Slide No	95	Slide No	96
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L20	Name	L21
Location	Near Kendri Vidyalaya, Halflong	Location	Near Kendri Vidyalaya, Halflong
NH/SH affected		NH/SH affected	
Lat(dd)	25.186756	Lat(dd)	25.186605
Lon(dd)	93.01923	Lon(dd)	93.01952
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583
Slide ID	field_visit/2022/Assam/97	Slide ID	field_visit/2022/Assam/98
Source	field_visit	Source	field_visit
Slide No	97	Slide No	98
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L23	Name	L24
Location	Near Kendri Vidyalaya, Halflong	Location	Near Kendri Vidyalaya, Halflong
NH/SH affected		NH/SH affected	
Lat(dd)	25.186619	Lat(dd)	25.18669
Lon(dd)	93.019455	Lon(dd)	93.01947
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583

Slide ID	field_visit/2022/Assam/99	Slide ID	field_visit/2022/Assam/100
Source	field_visit	Source	field_visit
Slide No	99	Slide No	100
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L25	Name	L26
Location	Near Kendri Vidyalaya, Halflong	Location	Near Kendri Vidyalaya, Halflong
NH/SH affected		NH/SH affected	
Lat(dd)	25.186378	Lat(dd)	25.186222
Lon(dd)	93.01897	Lon(dd)	93.01888
Date of Event	15-05-2022	Date of Event	15-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	1
Casualty	0	Casualty	0
Rainfall(mm)	183.10583	Rainfall(mm)	183.10583
Slide ID	field_visit/2022/Assam/101	Slide ID	Newspapers&Media/2022/Assam/102
Source	field_visit	Source	Newspapers&Media
Slide No	101	Slide No	102
State	Assam	State	Assam
District	Dima Hasao	District	Dima Hasao
Name	L30	Name	Hokai_pungchi
Location	Near Kendri Vidyalaya, Halflong	Location	Hokai Pungchi near Halflong
NH/SH affected		NH/SH affected	
Lat(dd)	25.186321	Lat(dd)	25.159405
Lon(dd)	93.02017	Lon(dd)	92.9836
Date of Event	15-05-2022	Date of Event	14-05-2022
Time of Event		Time of Event	
Type of Event	1	Type of Event	2
Casualty	0	Casualty	3
Rainfall(mm)	183.10583	Rainfall(mm)	182.39934
Slide ID	Newspapers&Media/2022/Assam/104	Slide ID	Newspapers&Media/2022/Assam/105
Source	Newspapers&Media	Source	Newspapers&Media
Slide No	104	Slide No	105
State	Assam	State	Assam
District	Cachar	District	Kamrup Metro
Name	Borakhai_tea_estate	Name	Navagarha_guard_wall_collapse
Location	Borakhai Tea Estate near NIT Silchar	Location	Chitrachal Hill, Navagraha
NH/SH affected		NH/SH affected	
Lat(dd)	24.759151	Lat(dd)	26.19286
Lon(dd)	92.7794	Lon(dd)	91.76223
Date of Event	18-06-2022	Date of Event	19-06-2022
Time of Event		Time of Event	
Type of Event	2	Type of Event	2
Casualty	2	Casualty	0
Rainfall(mm)	25.269623	Rainfall(mm)	48.64218

Slide ID	Newspapers&Media/2022/Assam/111	Slide ID	Newspapers&Media/2022/Assam/112
Source	Newspapers&Media	Source	Newspapers&Media
Slide No	111	Slide No	112
State	Assam	State	Assam
District	Kurung Kumey	District	Gopalpara
Name	Nizarpur_landslide	Name	Azadnagar_landslide
Location	Boragaon	Location	Azadnagar, Gopalpara
NH/SH affected		NH/SH affected	
Lat(dd)	26.11816	Lat(dd)	26.14118
Lon(dd)	91.683105	Lon(dd)	90.63741
Date of Event	12-06-2022	Date of Event	17-06-2022
Time of Event		Time of Event	
Type of Event	2	Type of Event	2
Casualty	4	Casualty	2
Rainfall(mm)	59.594055	Rainfall(mm)	101.55899

ARUNACHAL PRADESH

Slide ID	Newspapers&Media/2022/Arunachal Pradesh/106	Slide ID	Newspapers&Media/2022/Arunachal Pradesh/107
Source	Newspapers&Media	Source	Newspapers&Media
Slide No	106	Slide No	107
State	Arunachal Pradesh	State	Arunachal Pradesh
District	Papum Pare	District	Papum Pare
Name	Sood_landslide	Name	Punjabi_dhabha_landslide
Location	Sood Village near Yupia	Location	Behind punjabi dhabha sector-d itanagar
NH/SH affected	Yupia-Doimukh Road	NH/SH affected	
lat(dd)	27.112083	lat(dd)	27.095318
Lon(dd)	93.6878	Lon(dd)	93.62141
Date of Event	19-06-2022	Date of Event	16-05-2022
Time of Event	08:00	Time of Event	21:30
Type of Event	2	Type of Event	2
Casualty	1	Casualty	4
Rainfall(mm)	44.046043	Rainfall(mm)	53.276478

Slide ID	Newspapers&Media/2022/Arunachal Pradesh/108	Slide ID	Newspapers&Media/2022/Arunachal Pradesh/109
Source	Newspapers&Media	Source	Newspapers&Media
Slide No	108	Slide No	109
State	Arunachal Pradesh	State	Arunachal Pradesh
District	Papum Pare	District	Papum Pare
Name	Ganga_Jully_basti_road	Name	Modirijo_landslide
Location	NES secretariat Richi Jollang	Location	Shiv Mandir area, Modirijo Village between Itanagar and Naharlagun
NH/SH affected		NH/SH affected	NH 415
lat(dd)	27.079184	lat(dd)	27.107346
Lon(dd)	93.61592	Lon(dd)	93.650566
Date of Event	16-05-2022	Date of Event	19-06-2022
Time of Event		Time of Event	
Type of Event	2	Type of Event	2
Casualty	2	Casualty	0
Rainfall(mm)	53.276478	Rainfall(mm)	44.046043

Slide ID	Newspapers&Media/2022/Arunachal Pradesh/110
Source	Newspapers&Media
Slide No	110
State	Arunachal Pradesh
District	Papum Pare
Name	Sulung_tapin_landslide
Location	Koloriang circle
NH/SH affected	
lat(dd)	27.856197
Lon(dd)	93.27676
Date of Event	19-04-2022
Time of Event	
Type of Event	2
Casualty	3
Rainfall(mm)	0

MANIPUR

Slide ID	Newspapers&Media/2022/Manipur/120
Source	Newspapers&Media
Slide No	120
State	Manipur
District	Noney
Name	Tupul_yard_landslide
Location	Tupul Railway Yard Construction Camp
NH/SH affected	
Lat(dd)	24.80265
Lon(dd)	93.6709
Date of Event	30-06-2022
Time of Event	
Type of Event	2
Casualty	8
Rainfall(mm)	5.613702

MIZORAM

Slide ID	Newspapers&Media/2022/Mizoram/103
Source	Newspapers&Media
Slide No	103
State	Mizoram
District	Lunglei
Name	Tawipui_road
Location	Tawipui_road
NH/SH affected	Tawipui_road
Lat(dd)	22.690073
Lon(dd)	92.84086
Date of Event	19-06-2022
Time of Event	
Type of Event	2
Casualty	0
Rainfall(mm)	0.3301647

SIKKIM

Slide ID	Newspapers&Media/2022/Sikkim/121
Source	Newspapers&Media
Slide No	121
State	Sikkim
District	North Sikkim
Name	Poom_landslide
Location	Poom, Yumthang_Valley
NH/SH affected	Yumthang_valley_road
Lat(dd)	27.830094
Lon(dd)	88.695972
Date of Event	31-08-2022
Time of Event	16:00
Type of Event	2
Casualty	0
Rainfall(mm)	0.992669522
Slide ID	Newspapers&Media/2022/Sikkim/122
Source	Newspapers&Media
Slide No	122
State	Sikkim
District	South Sikkim
Name	Yangang Landslide
Location	Yangang
NH/SH affected	
Lat(dd)	27.296944
Lon(dd)	88.395556
Date of Event	04-11-2022
Time of Event	
Type of Event	2
Casualty	0
Rainfall(mm)	0.010780328

ANNEXURE-II: NEWSPAPER (CLIPPINGS & DIGITAL ARCHIVE)

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NEWS / SIKKIM / SIKKIM: LANDSLIDE HITS YUMTHANG VALLEY, BLOCKADE ON ROAD & COMMUNICATION

Sikkim: Landslide hits Yumthang valley; blockade on road & communication



Updated: Sep 01, 2022, 6:17 PM IST

A massive landslide hit Sikkim's Yumthang valley on August 31st around 4 pm in the evening. 70 tourists, who were stranded at Poom on the way to Yumthang in the northern part of Sikkim, have been rescued.

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At least six people were killed and several others feared missing after a massive landslide hit Tupul railway station in Manipur's Noney district.

Updated: June 30, 2022 5:42 PM IST

By India.com News Desk Edited by Surabhi Shourya



Massive Landslide in Tupul Rly Construction Camp in Manipur, 6 Dead, Over Dozens Injured

Imphal, Manipur: At least seven people were killed and several others feared missing after a massive landslide hit Tupul railway construction camp in Manipur's Noney district. The incident took place on Wednesday night, however, the identities of those killed in the incident are yet to be ascertained. Meanwhile, rescue operations by the Indian Army and Assam Rifles are

The Shillong Times ESTABLISHED 1945

MEGHALAYA NATIONAL INTERNATIONAL EDITORIAL COVID-19 REGIONAL SALANTINI JANERA TODAY

Home News Alert 3 die in landslide at Baghmara

By From Our Correspondent On Jun 17, 2022



Tura, June 17: Three persons were killed in a landslide at Baghmara near the Circuit House. The bodies have recovered.

Meanwhile, roads have been blocked because of inundation in South Garo Hills district where one bridge has collapsed in Rongara even as rivers are flowing above the danger level.

TODAY'S WEATHER

ST VIDEOS

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Omicron BF.7 threat Covid protocol

Agencies Jun 25, 2022

Bengaluru, Oct 25: Omicron sub-variant experts have

TS BREAKING Floods Flooding Assam: Around 2500 Arms Act Cases Registered Since 2015

Thirteen killed, one missing after rainfall-triggered landslides in Meghalaya

Meghalaya Chief Minister Conrad K Sangma took to the Twitter handle to offer his deepest condolences

Twitter Wednesday June 17, 2022 9:36 pm



Guwahati: At least thirteen persons including a Border Security Force (BSF) jawan have been killed in landslides triggered by incessant rains in Meghalaya in the last 24 hours.

In the East Khasi Hills district, at least seven individuals reportedly died due to landslides in Mawsonram Block, while, two others died in South West Khasi Hills.

One more person was reported missing at Kennymaw under Mawsonram Block and rescue efforts are underway, informed the chief minister of Meghalaya in a tweet.

Editorial

September 1, 2022

Three Years of Assam's NRC: A Tale of Uncertainty & Despair

In May 2021, Assam's State Coordinator of NRC again moved Supreme Court demanding a re-verification of the NRC published on August 31, 2019

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YOU ARE AT: English News India Heavy rainfall triggers landslide in Meghalaya

Heavy rainfall triggers landslide in Meghalaya

Meghalaya Chief Minister Conrad Sangma has announced an ex-gratia of Rs 4 lakh to the next kin of each of the deceased. Sangma also held a review meeting to assess the situation in the state's Garo Hills region which has been hit by landslides and floods last week.

Edited by: Paras Bisht @ParasBisht15 Shillong

Published on: June 17, 2022 14:56 IST



Image Source: TWITTER@ANI

Heavy rainfall triggers landslide in Meghalaya

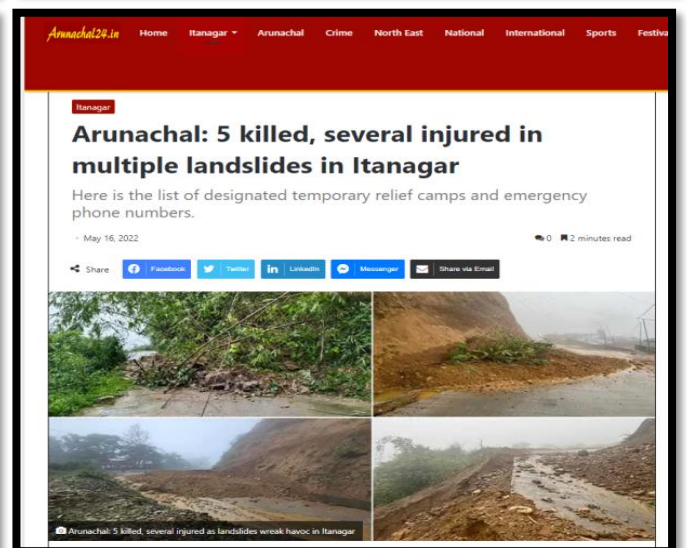
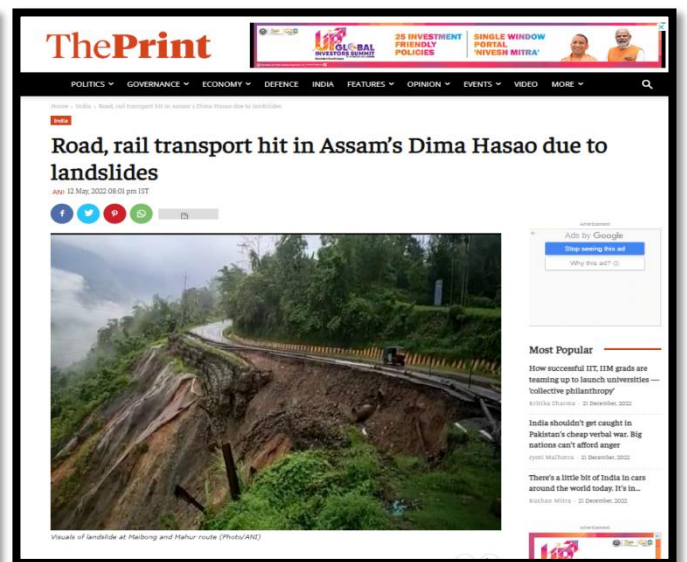
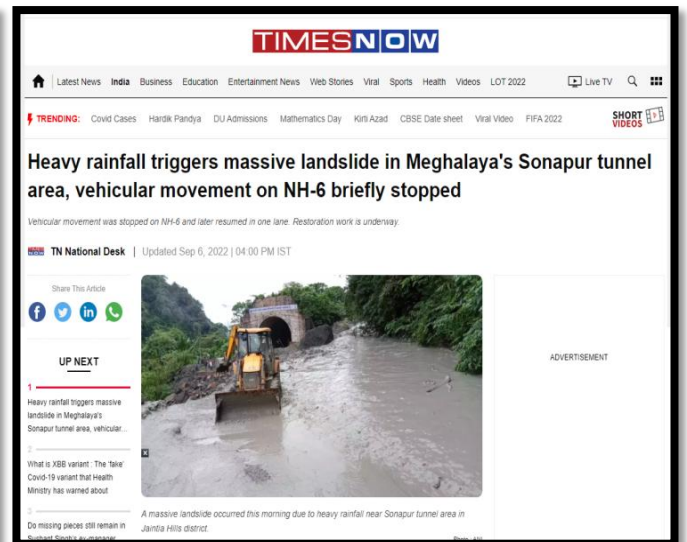
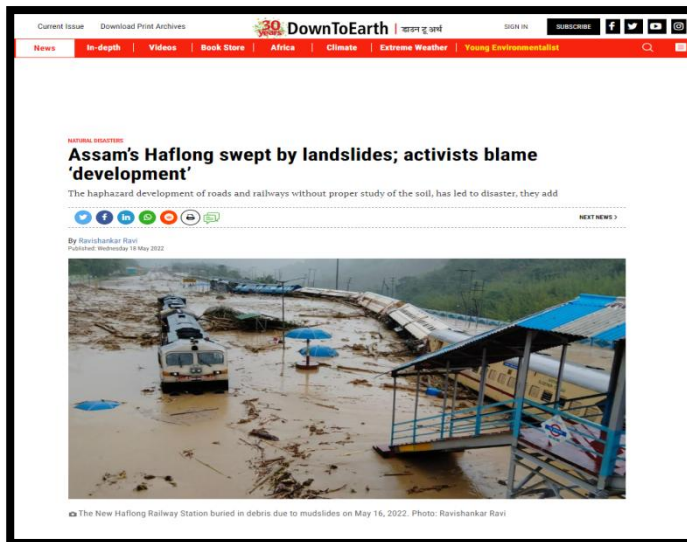
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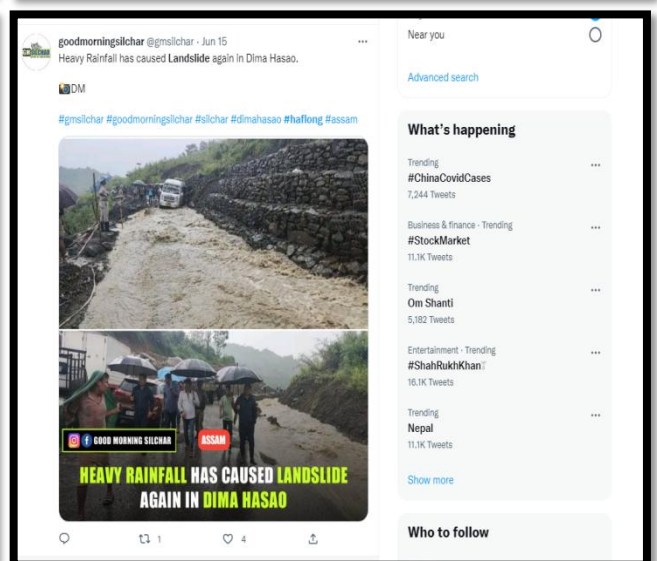
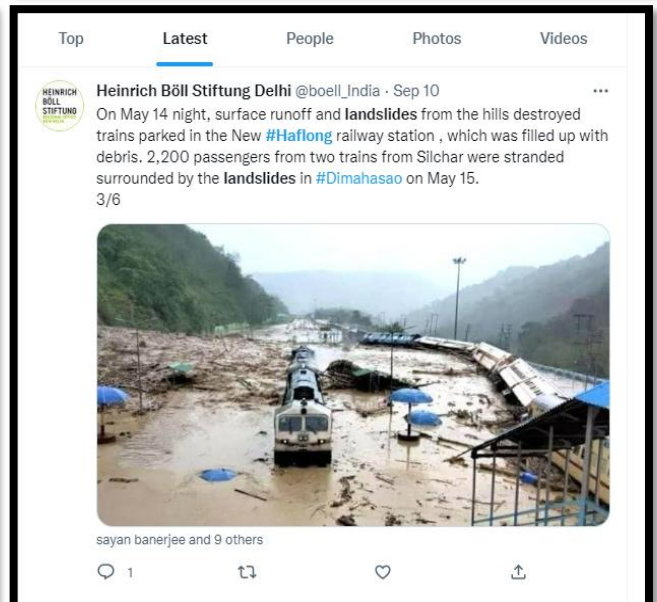
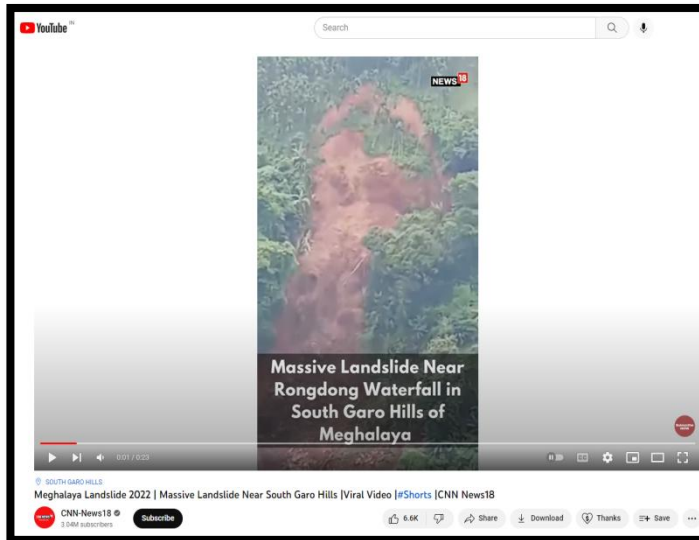
TOP NEWS

Unfortunate: Dhankhar on Sonia Gandhi's 'govt seeking to delegitimise judiciary' remarks

Varun Sood breaks silence on ex-girlfriend Divya Agarwal's engagement: 'It was not hinted..'



ANNEXURE-III: ONLINE MEDIA



ANNEXURE-IV: INVENTORY ATTRIBUTES

Slide ID	Unique Identifier for Landslide
Source	Newspaper/Media, SDMA, GSI or From Field
Slide No	Serial number
State	State in NER where landslide occurred
District	District where landslide occurred
Name	Name of the event
Location	Specific location from SDMA reports. General location – neighbourhood, village name etc from newspapers
NH/SH Cutoff/Affected	If National Highway, State Highway or any other important road was cutoff or affected
Latitude	Exact Latitude from GSI/SDMA reports General Latitude from Newspaper Articles
Longitude	Exact Longitude from GSI/SDMA reports General Longitude from Newspaper Articles
Date of Event	Date of the event
Time of Event	Time of event wherever possible
Type of Event	Type 1, 2 or 3 (explained in the next section)
Casualty	Casualty figures
Rainfall (24 hrs)	24 hours rainfall prior to the event

ANNEXURE-V: CODE SAMPLE

```
*C:\Users\NESAC\Desktop\2020\python\gee_code.py - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

gee_code.py
1 import ee
2 from ee_plugin import Map
3
4 #Code for displaying Pre and Post Imagery
5
6 # Processing polygon_ls
7 polygon_ls = "users/tusharmontfort/senapati"
8
9 aoi = ee.FeatureCollection(polygon_ls);
10
11 # ENTER THE TIME FRAME for the first image
12 start_date1 = '2020-10-01'
13 end_date1 = '2020-10-23'
14
15
16
17 # Import the Sentinel 2 collection as sentinel 2
18 sentinel2_1 = ee.ImageCollection("COPERNICUS/S2")
19
20 # Import the boundary
21 ls_bnd = ee.FeatureCollection(polygon_ls)
22
23 # Filter the image collection using filterBounds() and filterDate() method.
24 # Sort the collection by cloud cover metadata
25 image1 = sentinel2_1 \
26     .filterBounds(ls_bnd) \
27     .filterDate(start_date1, end_date1) \
28     .sort('CLOUDY_PIXEL_PERCENTAGE', False) \
29     .map(lambda image1: image1.addBands(image1.metadata('system:time_start')))\
30     .mosaic()\
31     .clip(ls_bnd)
32
33
34 #TRUE COLOUR COMPOSITE
35 visualization_TCC = {
36     'min': 0.0,
37     'max': 3000,
38     'bands': ['B4', 'B3', 'B2']
39 };
40
41 # FALSE COLOUR COMPOSITE
42 visualization_FCC = {
43     'min': 0.0,
44     'max': 3000,
45     'bands': ['B8', 'B4', 'B3']
46 };
47
48 LS_Scene = ee.Image(image1).select(['B4', 'B3', 'B2', 'B8']);
49
50 Map.addLayer(LS_Scene, visualization_TCC, 'true_color_composite_pre');
51 Map.addLayer(LS_Scene, visualization_FCC, 'False_color_composite_pre');
```

Note – This report does not include all landslides in North Eastern Region for the year 2022.